

A Modular General-Purpose Semi-Humanoid Robotic Platform

Mechatronic System Design and Conceptual Architecture

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Abstract

Semi-humanoid robotics unites the mobility of wheeled autonomous mobile robots (AMRs) with the dexterous reaching and manipulation capabilities of dual-arm anthropomorphic upper bodies. This report establishes the conceptual mechatronic system design and foundational architecture for a modular, general-purpose semi-humanoid robotic platform engineered in accordance with the VDI 2206 design methodology for mechatronic and cyber-physical systems. Grounded in standardized service robotics benchmarks (RoboCup@Home) and contemporary morphological analyses, the platform pairs a high-traction differential drive mobile base (2 active 200 mm rubber wheels + 2 passive swivel casters) with a rigid structural mast torso (vertical reach envelope 500–1350 mm; with an optional modular telescopic column analyzed and excluded from the baseline build), dual 6-DoF Quasi-Direct Drive (QDD) manipulators, compliant 2-finger parallel-jaw end effectors, and an expressive 2-DoF stereo-vision perception head, establishing an integrated kinematic chain of 18 active degrees of freedom. Crucially, the system formalizes a strict decoupling between hardware capability and application intent through a standardized Task Port abstraction layer. The report details the complete engineering requirements list (*Anforderungsliste*), functional flow structures (material, energy, and information), hierarchical multi-rate software stack (from 1 kHz real-time fieldbuses to Vision-Language-Action embodied AI models), modular device-level subsystem decompositions, cross-domain engineering allocations, Bill of Materials (BOM) estimates, and a phased implementation roadmap.

Keywords: semi-humanoid robotics; mechatronic system design; VDI 2206; mobile manipulation; modular architecture; task port abstraction; Vision-Language-Action (VLA); ROS 2; differential drive.

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1 Introduction and Methodological Framework

1.1 Context and Motivation: The Spectrum of Robotic Automation

The contemporary landscape of robotics and autonomous systems exhibits a profound bifurcation between specialized, rigid industrial automation on one extreme and full bipedal humanoid platforms on the other. For decades, conventional industrial automation has achieved unmatched throughput, repeatability, and structural rigidity within highly structured manufacturing cells. However, these traditional manipulators and Automated Guided Vehicles (AGVs) rely strictly on engineered environmental fixtures, predictable workpieces, and physical safety fencing that isolates machinery from human workers. In unstructured, human-centric environments—such as domestic residences, eldercare and clinical facilities, research laboratories, and agile light-industrial logistics hubs—such fixed automation is fundamentally non-viable due to its complete inability to adapt to geometric uncertainties, variable lighting, dynamic obstacles, and unexpected physical contact.

Conversely, the robotics research community and emerging commercial ventures have increasingly pursued full bipedal humanoids. While anthropomorphic bipedal morphologies represent the theoretical ideal for traversing arbitrary human architectural features (such as multi-flight staircases and narrow step-overs), they introduce formidable mechatronic penalties:

1. **High Dynamic Instability and Fall Hazards:** Bipedal locomotion is inherently underactuated and governed by non-linear inverted-pendulum dynamics. Maintaining dynamic balance in populated or cluttered environments presents severe risks; a catastrophic loss of balance results in high-energy kinetic impacts that jeopardize human safety and inflict severe structural damage upon the robot itself.
2. **Excessive Energetic Overhead:** Bipedal systems consume substantial electrical power merely to maintain static equilibrium and vertical posture against gravitational acceleration ($\sim 200\text{--}500$ W baseline quiescent draw), drastically curtailing battery endurance down to 1.0–2.0 hours under continuous operation.
3. **Actuator Complexity and Thermal Limitations:** High-torque-density actuators required for explosive leg dynamics generate extreme internal thermal flux, requiring complex liquid or active convective cooling systems.
4. **Prohibitive Cost and Maintenance Overhead:** Capital expenditure for research-grade bipedal platforms typically exceeds \$100,000–\$250,000, with fragile harmonic or planetary drivetrains that suffer frequent mechanical degradation during foot-ground impact events.

To bridge this operational and economic chasm, semi-humanoid robotic platforms have emerged as a pragmatically superior architectural paradigm [1, 2, 3, 4]. A semi-humanoid robot synthesizes the proven stability, energy efficiency, and high payload-to-weight ratio of a wheeled mobile base with the functional versatility and dexterity of an anthropomorphic upper body. By integrating a high-traction differential mobile base (2 active rubber wheels + 2 passive swivel casters) with a rigid structural mast torso, dual articulated 6-Degree-of-Freedom (DoF) manipulators, compliant 2-finger parallel-jaw end effectors, and a sensorized 2-DoF pan-tilt perception head, semi-humanoids unlock natural human-scale spatial reaching (500–1350 mm vertical reach envelope) and bimanual object manipulation across an integrated chain of 18 active degrees of freedom. Crucially, wheeled locomotion eliminates the threat of dynamic tipping during stationary manipulation, reduces energetic expenditure during idle co-presence to a nominal baseline, and provides a dependable, high-payload mobile foundation for long-horizon autonomous tasks.

1.2 The Semi-Humanoid Platform as a Cyber-Physical System for Industry 5.0

The conceptualization and embodiment of this semi-humanoid platform is directly aligned with the foundational principles of Industry 5.0. Whereas the preceding Industry 4.0 paradigm concentrated primarily on hyper-connectivity, end-to-end digitisation, continuous telemetry extraction, and automated “lights-out” productivity,

Industry 5.0 places human-centricity, safe physical co-presence, environmental resilience, and sustainability at the core of technological development.

Under Industry 5.0, autonomous robots are no longer relegated to fenced enclosures or treated as isolated mechanical agents; rather, they serve as collaborative partners operating within shared workspaces alongside human operators. Within this framework, the semi-humanoid platform operates as a multi-tier Cyber-Physical System (CPS), defined by a seamless, bi-directional coupling between discrete computational algorithms and continuous physical domain dynamics:

- **Physical Layer (Mechatronic Embodiment):** Comprises structural linkages, multi-axis Quasi-Direct Drive (QDD) actuators, high-capacity lithium energy storage, exteroceptive multimodal sensors (3D LiDAR, RGB-D vision, microphone arrays), and proprioceptive feedback elements (optical encoders, motor phase current sensors, multi-axis force-torque transducers).
- **Cyber Layer (Embedded & Cognitive Stack):** Comprises distributed real-time microcontrollers executing closed-loop fieldbus control at 1 kHz, heterogeneous System-on-Chip (SoC) edge computers hosting autonomous Simultaneous Localization and Mapping (SLAM), kinematic trajectory generation, behavior-tree execution, and embodied Vision-Language-Action (VLA) foundation models.
- **Safe Co-Presence and Resilient Interaction:** Certified compliance with personal care robot safety mandates (specifically ISO 13482:2014 [5]), governed by deterministic hardware safety interlocks, contact force limitation ($F_{\text{contact}} \leq 50\text{--}150\text{ N}$ depending on contact geometry), active software speed-and-separation monitoring, and transparent backdrivability to absorb unexpected kinetic disturbances without inflicting injury or sustaining mechanical damage.

1.3 Methodological Framework: The VDI 2206 Standard

Developing an advanced multi-DoF mobile manipulator demands an integrated, concurrent engineering methodology. Traditional sequential development models—wherein mechanical structures are drafted in isolation, followed sequentially by electrical wiring, motor selection, and ad-hoc software patching—consistently induce structural resonance mismatches, thermal throttling, excessive parasitic mass, and communication bus saturation.

To guarantee architectural coherence and eliminate domain interface mismatches, this project formally adopts the VDI 2206 design methodology (“Design Methodology for Mechatronic Systems and Cyber-Physical Systems”) [6]. The VDI 2206 framework structures development across two coupled operational scales: the overarching *Macro V-Model* and the recurring *Micro-Cycle of Problem Solving*.

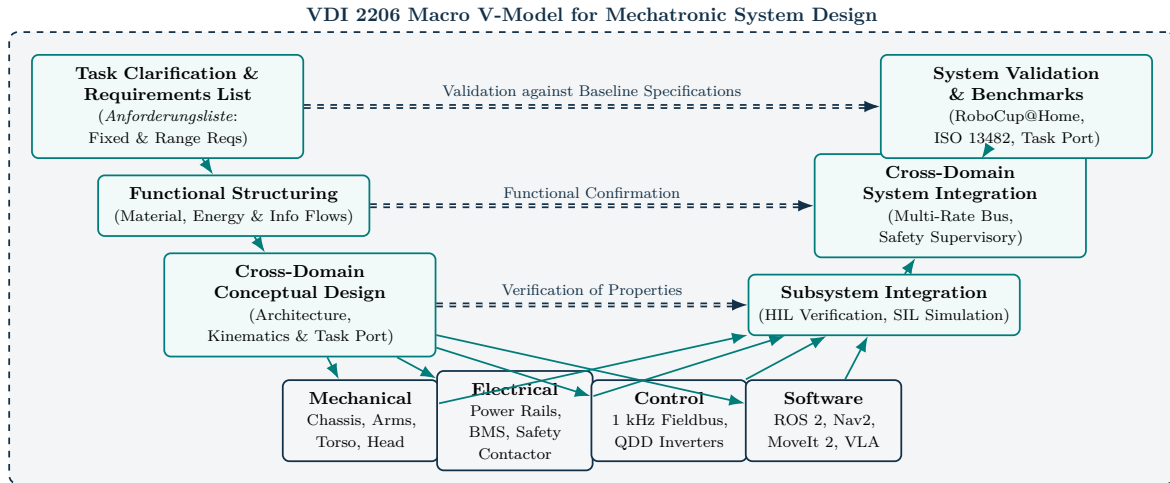


Figure 1: The VDI 2206 macro-level V-model adapted for the modular semi-humanoid platform, illustrating the descent from requirements formulation to domain-specific elaboration, followed by ascending multi-stage verification and validation.

1.3.1 The Macro V-Model

As illustrated in Figure 1, the macro V-model coordinates the overall lifecycle across three primary phases:

1. **Left Descending Branch (Specification & Conceptual Design):** Begins with comprehensive task clarification and drafting a structured requirements list (*Anforderungsliste*), dividing specifications into fixed requirements (*Festforderungen*) and range requirements (*Bereichsforderungen*). These specifications are abstracted into functional structures modeling conserved transport and transformation of *material*, *energy*, and *information*. From these abstractions, cross-domain architectures and kinematics are synthesized.
2. **Base of the V-Model (Domain-Specific Concurrent Elaboration):** Development branches concurrently into mechanical structural design (FEA optimization, link geometry, drivetrain selection), electrical hardware design (power architecture, galvanic isolation, BMS), embedded control design (real-time motor commutation, fieldbus deterministic timing), and computer software development (ROS 2 nodes, kinematic solvers, behavior trees).
3. **Right Ascending Branch (Integration, Verification, and Validation):** Concurrently developed domain models and physical sub-assemblies are integrated hierarchically. Physical components undergo Software-in-the-Loop (SIL) simulation in high-fidelity physics engines, followed by Hardware-in-the-Loop (HIL) dynamometer testing, full mechatronic integration, and empirical validation against standardized benchmark scenarios.

1.3.2 The Recurring Problem-Solving Micro-Cycle

Operating recursively within every macro-phase of the V-model is the 6-stage VDI 2206 problem-solving micro-cycle. This systematic cognitive loop compels the engineering team to validate conceptual variants objectively before freezing interfaces:

1. **Situation Analysis (*Situationsanalyse*):** Rigorous audit of current boundary conditions, state-of-the-art benchmarks (e.g., RoboCup@Home, TIAGo Pro, Reachy 2), physical bottlenecks, and failure modes observed in existing platforms.
2. **Goal Formulation (*Zielformulierung*):** Definition of unambiguous, quantitative target criteria (e.g., arm settling time $t_{\text{settling}} \leq 0.4$ s, continuous payload ≥ 2.0 kg, footprint ≤ 540 mm).
3. **Solution Synthesis (*Synthese von Lösungen*):** Generation of multiple divergent morphological concepts without premature design lock-in (e.g., contrasting prismatic elevating torso columns against revolute waist pitch joints).
4. **Analysis and Evaluation (*Analyse und Bewertung*):** Quantitative trade-off analysis utilizing multi-criteria decision matrices, kinematic simulation, static/dynamic FEA, and bus latency modeling.
5. **Decision (*Entscheidung*):** Selection of the optimal concept based on engineering metrics, manufacturing feasibility, cost, and reliability.
6. **Planning (*Planung*):** Structuring detailed work packages, defining physical/logical interface contracts, and scheduling verification milestones for the subsequent development cycle.

1.4 Core Mechatronic Dynamic Principles

High-performance mobile manipulation requires an uncompromising focus on structural mechanics and dynamic control bandwidth. Unlike static industrial manipulators anchored to rigid concrete foundations, a mobile manipulator operates atop a movable chassis with finite compliance at the wheel-ground contact interface. Rapid arm accelerations generate reaction forces and moments that can induce body sway, structural vibrations, and end-effector tracking errors.

1.4.1 Structural Natural Frequency and Resonance

To guarantee responsive, vibration-free motion during high-speed reaching and trajectory tracking, each actuated mechanical link and joint assembly is modeled to first order as an equivalent spring-mass-damper system. The undamped natural angular frequency ω_n of the joint assembly is governed by:

$$\omega_n = \sqrt{\frac{K}{M}} \quad (1)$$

where K represents the aggregate mechanical stiffness of the drivetrain (incorporating gearbox torsional stiffness, bearing rigidity, and structural link flexure in $\text{N} \cdot \text{m}/\text{rad}$), and M denotes the effective moving mass or reflected rotational inertia about the joint axis in $\text{kg} \cdot \text{m}^2$.

1.4.2 Settling Time and Operational Bandwidth

Following a rapid point-to-point repositioning step command, the settling time t_{settling} (defined as the duration required for Cartesian oscillations to decay permanently within a specified 2% error band) is approximated by:

$$t_{\text{settling}} \approx \frac{4}{\zeta \omega_n} \quad (2)$$

where ζ represents the effective damping ratio of the closed-loop joint assembly. For critically damped or moderately underdamped systems ($0.65 \leq \zeta \leq 0.85$), minimizing settling time directly demands maximizing the natural frequency ω_n . Low natural frequencies ($\omega_n < 10$ Hz) result in prolonged structural ringing, severely throttling closed-loop control gains and slowing autonomous pick-and-place cycles.

1.4.3 Mass Minimization of Distal Links

Equation (1) indicates that maximizing ω_n requires either increasing stiffness K or decreasing moving inertia M . However, adding physical material to increase stiffness K inherently increases mass M , often degrading performance.

In articulated serial kinematic chains, the total kinetic energy and reflected inertia at proximal actuators are critically dependent on the spatial distribution of mass. By the parallel axis theorem and manipulator Jacobian inertia transformation, the effective inertia $I_{\text{reflected},j}$ reflected across joint j due to distal links $i > j$ scales quadratically with link reach:

$$I_{\text{reflected},j} = \sum_{i=j}^N (I_i + m_i \mathbf{r}_{j,i}^T \mathbf{r}_{j,i}) \quad (3)$$

where m_i is the mass of link i , I_i is its centroidal inertia tensor, and $\mathbf{r}_{j,i}$ is the position vector from joint axis j to the center of mass of link i .

Consequently, minimizing the mass of distal components (such as wrist pitch/roll mechanisms and end-effector grippers) produces a quadratic-to-cubic reduction in reflected inertia at shoulder and elbow joints. This mass reduction shifts structural resonance well above the operational control bandwidth ($\omega_n \geq 25\text{--}35$ Hz), suppressing oscillations and permitting aggressive, high-bandwidth closed-loop trajectory tracking.

1.4.4 Structural Rigidity versus Compliance: The Quasi-Direct Drive Paradigm

A classical conflict in robotics engineering exists between structural rigidity and mechanical compliance:

- **Rigid Actuation with High-Ratio Reductions** ($N > 100 : 1$): Utilizes multi-stage planetary or harmonic gearboxes to achieve high static holding torque in compact envelopes. However, reflected rotor inertia scales with the square of the gear ratio ($I_{\text{reflected}} = N^2 I_{\text{rotor}}$), resulting in massive mechanical impedance, extreme friction, non-linear backlash, and zero backdrivability. In collaborative environments, accidental physical collisions impart destructive shock loads directly onto the gear teeth, risking mechanical failure and causing severe human injury.
- **Series Elastic Actuators (SEAs)**: Introduce physical springs between the gearbox and the load to measure torque via deflection and provide passive impact absorption. However, the intentional introduction of mechanical compliance drastically limits control bandwidth, introduces low-frequency structural resonances ($\omega_n < 8$ Hz), and degrades high-speed Cartesian precision.
- **Quasi-Direct Drive (QDD) Solution**: To reconcile these conflicting demands, our platform adopts Quasi-Direct Drive (QDD) actuators across the arm joints. QDD pairs high-torque-density, multi-pole brushless DC outrunner motors with single-stage, low-ratio planetary gearings ($1 : 6 \leq N \leq 1 : 10$). This low transmission ratio preserves ultra-low reflected inertia ($I_{\text{ref}} \sim 36\text{--}100 \times I_{\text{rotor}}$), delivers transparent mechanical backdrivability, absorbs physical impact shocks without gear stripping, and enables direct, high-fidelity joint torque estimation via motor phase current feedback:

$$\tau_{\text{joint}} \approx N \cdot K_{\tau} \cdot I_q \quad (4)$$

where K_{τ} is the motor torque constant and I_q is the quadrature field-oriented control (FOC) current. This allows the software stack to execute active impedance control, presenting soft, compliant virtual stiffness to human co-workers while maintaining high structural rigidity during autonomous manipulation.

1.5 Core Architectural Philosophy: Separation of Capability from Intent

The central engineering thesis of this project is the rigorous separation of *capability* (the physical robotic platform) from *intent* (the downstream user application or embodied AI cognitive agent).

Autonomous mobile manipulation robots deployed in service and research settings are overwhelmingly engineered as specialized, single-purpose devices: a rigid arm permanently attached to a custom chassis, driven by monolithic software where high-level task logic is entangled directly with motor PWM drivers and kinematics. When the target task changes, the entire system must be fundamentally redesigned.

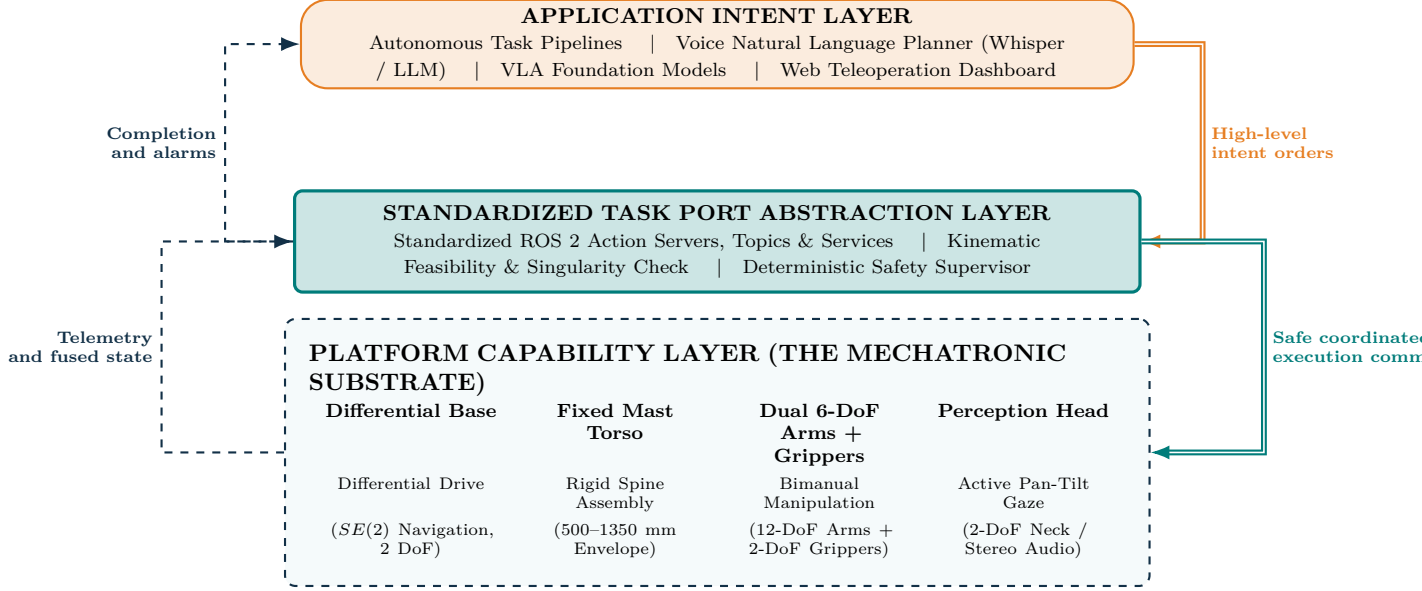


Figure 2: The foundational architectural separation between application intent and platform capability mediated through the standardized Task Port abstraction layer.

This project establishes the exact opposite paradigm. The robotic platform is engineered as a general-purpose, modular mechatronic substrate that is completely agnostic to the specific downstream application. What the platform physically possesses—a high-traction differential mobile chassis that navigates collision-free, a rigid structural mast torso maintaining a 500–1350 mm ergonomic reaching envelope, dual 6-DoF QDD arms with compliant parallel-jaw grippers, and a sensorized 2-DoF pan-tilt head that tracks objects and human dialogue (establishing 18 active DoF total)—remains invariant.

As diagrammed in Figure 2, the user or cognitive reasoning agent never accesses low-level joint motor drivers, fieldbus packet frames, or kinematic inversion matrices. Instead, all interaction occurs through a formal, standardized abstraction layer termed the **Task Port**. The application layer supplies *intent* (e.g., “Pick up the medication bottle from the kitchen counter and bring it to the dining table”), while the platform orchestrates *capability*—autonomously handling perception, global path planning, collision-free inverse kinematics, multi-rate trajectory interpolation, and active contact safety.

1.6 Document Objectives and Outline

This document establishes the comprehensive mechatronic system design, architectural specifications, and engineering allocations for the proposed modular semi-humanoid robot. The remainder of this report is organized as follows:

- Section 2 grounds platform capabilities in empirical service robotics benchmarks (RoboCup@Home, World Robot Summit, ISO 13482), extracts architectural boundary constraints, conducts morphological benchmarking against contemporary platforms, and defines concrete validation tasks.
- Section 3 establishes the formal engineering requirements list (*Anforderungsliste*) classifying constraints across geometry, kinematics, dynamics, electrical power, embedded control, and software.
- Section 4 details the formal functional structure through conserved material, energy, and information flow models.

- Section 5 delineates the multi-tier system architecture, real-time communication fieldbuses, safety supervisor, and the Task Port API.
- Section 6 executes device-level deep dives into each functional mechatronic subsystem (mobile base, torso, arms, head, HMI, and power).
- Section 7 allocates multidisciplinary engineering tasks across mechanical, electrical, embedded firmware, and software domains.
- Section 8 details the preliminary Bill of Materials (BOM), budget allocation, and phased development roadmap.
- Section 9 synthesizes key mechatronic architectural takeaways and establishes next steps toward physical embodiment and testing.

2 Empirical Task Formulation and Benchmark Derivation

2.1 Grounding Operational Objectives in Standardized Robotics Benchmarks

Engineering a general-purpose robotic platform requires establishing operational targets that reflect the physical and cognitive challenges of real-world deployment. In academic literature, robotics projects frequently falter by engineering custom mechanisms tailored to arbitrary, isolated laboratory tasks (e.g., grasping a single specific cylinder on an uncluttered bench). When transferred to unconstrained human environments, such bespoke systems fail due to inadequate kinematic workspace, insufficient payload margins, rigid actuation, or brittle perceptual assumptions.

To ground the platform’s functional specifications in validated real-world requirements, this project derives its operational primitives from international competitive service robotics leagues and certified industrial safety standards. These benchmarks capture the environmental variability, clutter, geometric diversity, and dynamic human interactions inherent to human-inhabited spaces.

2.1.1 The RoboCup@Home Benchmark Suite

The international RoboCup@Home competition represents the premier global benchmark for autonomous service and domestic robotics [7]. Spanning the Domestic Standard Platform League (DSPL), Open Platform League (OPL), and Social Standard Platform League (SSPL), RoboCup@Home assesses mobile manipulators within realistic, fully furnished domestic apartments and office environments. Analyzing rulebooks, technical challenge papers, and operational logs reveals four canonical benchmark tasks that establish the minimum functional envelope of our platform:

1. **General Purpose Service Robot (GPSR):** The GPSR benchmark assesses physical and cognitive generalization without prior environment tuning. High-level commands are generated dynamically via formal grammar engines, categorised into three escalating complexity tiers:
 - *Category I (Fully Specified Tasks):* The command explicitly states the required action sequence, target object, and destination (e.g., “Navigate to the dining table, grasp the green soda can, and place it into the waste basket”). This tier benchmarks deterministic pipeline execution: autonomous Simultaneous Localization and Mapping (SLAM), 3D point-cloud object segmentation, inverse kinematic reachability, and grasp execution.
 - *Category II (Contextual Tasks):* The command contains ambiguous or missing spatial or semantic information (e.g., “Bring me a beverage from the kitchen”). The platform must perform semantic spatial reasoning, inspect candidate storage locations across multiple shelf tiers (300–1350 mm), and initiate spoken clarification dialogue through speech synthesis and directional microphone arrays.
 - *Category III (Erroneous and Contradictory Tasks):* The command involves physical impossibilities, missing items, or obstructed paths (e.g., requesting an item not present in the room). The robot must verify scene state discrepancies, detect tactile or visual grasp failures, abort hazardous actions safely, verbally articulate the failure mode to human users, and execute fallback behaviors.
2. **Clean Up and Table Clearing Benchmark:** This task formalizes physical environment rearrangement over the special Euclidean group $SE(3) = \mathbb{R}^3 \times SO(3)$. The autonomous agent must transform an unorganized environment from an initial state $s_0 \in \mathcal{S}$ to a desired tidy configuration $s^* \in \mathcal{S}$ through direct

physical contact. The robot must approach standard dining surfaces, segment cluttered tableware (cups, ceramic bowls, silverware, beverage bottles), estimate surface normals, and execute multi-angle grasps without toppling adjacent fragile objects. Furthermore, clearing tableware requires transporting items and placing them into confined dishwasher racks or designated storage bins.

3. **Interaction with Articulated Domestic Fixtures:** Service environments demand interaction with kinematically constrained fixtures. The robot must approach, unlatch, open, and close hinged cupboard doors, sliding drawers, and spring-loaded refrigerator doors. These operations impose closed kinematic chain constraints where the end-effector trajectory $\mathbf{x}_{EE}(t)$ is constrained along a 1-DoF circular arc or prismatic rail, requiring compliant contact force regulation ($F_{\text{contact}} \leq 30\text{--}50\text{ N}$) to prevent excessive internal stresses.
4. **Receptionist, Person Following, and Physical Object Handover:** This benchmark evaluates social co-presence and continuous physical human-robot interaction (pHRI). The platform must detect human poses, identify unfamiliar individuals, maintain continuous visual tracking during guided navigation ($v_{\text{walk}} \leq 1.2\text{ m/s}$), and preserve proxemic personal zones (0.8–1.5 m). During physical handovers (delivering a beverage or taking a carry bag), the robot must dynamically adjust arm trajectory to the human hand pose, detect contact via wrist force-torque sensing or motor current monitoring, and execute smooth, intentional payload releases at safe approach velocities ($v_{EE} \leq 0.3\text{ m/s}$).

2.1.2 World Robot Summit (WRS) Partner Robot Challenge

The World Robot Summit (WRS) Partner Robot Challenge [8] provides complementary empirical validation protocols centered on daily life assistance in standard living spaces. The WRS benchmark emphasizes:

- **Tidy-Up Operations in Cluttered Living Rooms:** Retrieval of deformable objects (clothing, blankets), thin items lying flat on the floor (pens, paper sheets), and small accessories stored inside low drawers (200–400 mm height).
- **Continuous Execution Autonomy:** Penalizing teleoperation interventions and rewarding continuous, uninterrupted multi-step task execution over extended time horizons (15–30 minutes).
- **Strict Safety Penalties:** Automatic disqualification or severe scoring deductions for collisions with furniture walls exceeding dynamic impact thresholds ($> 20\text{ N}$).

2.1.3 Safety Certification Standards: ISO 13482:2014

Deploying mobile manipulators in human-inhabited environments requires rigorous adherence to safety standards. The platform is engineered to comply with **ISO 13482:2014** (“Robots and robotic devices — Safety requirements for personal care robots”) [5], under the classification of a *mobile servant robot*. Key mandates governing the mechatronic design include:

- **Hazard Mitigation and Risk Reduction:** Systematic elimination of mechanical pinch and shear points at articulated arm joints, torso elevation columns, and wheel cowlings.
- **Emergency Stop and Operational Interlocks:** Integration of a Category 3, Performance Level d (PL d) dual-channel safety circuit capable of de-energizing all actuator drive inverters within $t_{\text{stop}} \leq 10\text{ ms}$ upon E-stop actuation or safety LiDAR field breach.
- **Limitation of Contact Forces and Speed:** Adherence to collaborative power and force limiting principles (incorporating ISO/TS 15066 guidelines), ensuring transient impact forces remain below human pain thresholds ($F_{\text{impact}} \leq 150\text{ N}$, quasi-static crushing force $F_{\text{crush}} \leq 50\text{--}75\text{ N}$), supported by backdrivable Quasi-Direct Drive (QDD) actuators and soft-skinned compliant end effectors.
- **Overturning Stability:** Maintaining static and dynamic stability under maximum arm reach ($R_{\text{arm}} \approx 750\text{ mm}$), maximum continuous payload ($m_{\text{load}} = 3.0\text{ kg per arm}$), and maximum base deceleration ($a_{\text{base}} = 1.0\text{ m/s}^2$).

Table 1: Systematic derivation of mechatronic primitives and physical constraints from empirical benchmarks.

Benchmark Source	Operational Sub-Tasks	Underlying Mechatronic Primitives	Physical & Environmental Constraints
RoboCup@Home GPSR (Cat. I) [7]	Autonomous pick-and-place, point-to-point transit, object sorting	Holonomic SLAM navigation, 3D point-cloud registration, $SE(3)$ trajectory generation, parallel grasp	Doorway width ≥ 750 mm; table height 720–750 mm; item mass ≤ 1.5 kg
RoboCup@Home GPSR (Cat. II) [7]	Contextual item retrieval, dialogue clarification, shelf search	Directional audio parsing (STT), semantic mapping, active pan-tilt visual search, multi-height reach	Ambient acoustic noise 50–65 dB; multi-tier shelf search (300–1350 mm)
RoboCup@Home GPSR (Cat. III) [7]	Discrepancy detection, grasp slip detection, safety abort	Proprioceptive current monitoring, wrist F/T tactile sensing, speech synthesis (TTS), exception recovery	Dynamic human obstacles; safe deceleration $a_{\text{dec}} \leq 1.0$ m/s ² ; timeout ≤ 5 min
Clean Up & Table Clearing [7]	Tableware segmentation, dishwasher loading, surface clearing	Dense RGB-D normal estimation, compliant grasp closure, high-accuracy Cartesian positioning ($\leq \pm 2$ mm)	Fragile ceramic/glass targets; grasp span 20–90 mm; vertical clearance ≤ 350 mm
WRS Partner Robot [8]	Tidy-up from floor, articulated drawer/door opening	Ground-level reaching ($z = 0$ mm), constrained circular/prismatic path following, impedance control	Floor-level grasping; pulling forces up to 40 N; zero wall collision damage
Receptionist & Handover [7]	Person following, proxemic navigation, compliant physical handover	2D/3D human skeletal tracking, dynamic costmap avoidance, soft wrist impedance, release timing	Human walking speed ≤ 1.2 m/s; social distance 0.8–1.5 m; handover speed ≤ 0.3 m/s
ISO 13482 Compliance [5]	Safe co-presence, emergency stop, physical impact limitation	Dual-channel hardware E-stop, software safety supervisory monitor, backdrivable QDD actuation	Safety cutoff $t_{\text{stop}} \leq 10$ ms; impact force $F \leq 150$ N; crushing force ≤ 50 N

2.2 Physical Boundary Conditions and Kinematic Sizing

Synthesizing the empirical requirements from Table 1 directly dictates the geometric envelope and kinematic reach of the semi-humanoid platform:

- **Doorway Clearances and Base Footprint:** Standard residential and commercial interior doorways exhibit clear aperture widths between 750 mm and 900 mm. To guarantee safe, collision-free transit with an adequate lateral clearance margin (≥ 75 –100 mm per side), the maximum mobile base footprint diameter is constrained to:

$$D_{\text{base}} \leq 540\text{--}600 \text{ mm} \quad (5)$$

- **Vertical Workspace Envelope:** Standard architectural furnishings exhibit strictly standardized working heights:
 - *Floor Level* ($z = 0 \text{ mm}$): Fallen objects, shoes, floor tidy-up.
 - *Dining Tables and Desks* ($z = 720\text{--}750 \text{ mm}$): Standard meal surfaces and workstations.
 - *Kitchen Counters and Workbenches* ($z = 850\text{--}900 \text{ mm}$): Food preparation, appliance manipulation.
 - *Intermediate and High Shelves* ($z = 1000\text{--}1350 \text{ mm}$): Cupboards, pantry shelves, wall cabinets.

By establishing the shoulder yoke at an ergonomic height of $\approx 950 \text{ mm}$ on a rigid structural mast, the dual 6-DoF manipulators cover a continuous vertical manipulation reach band from 500 mm to 1350 mm. This directly spans dining tables (750 mm), kitchen counters (900 mm), and eye-level shelves purely through joint articulation, avoiding the parasitic dead weight, mechanical binding risk, and cost of an active prismatic mast in the baseline build (while supporting an optional modular telescopic column for extended floor-to-ceiling reach).

- **Continuous Payload Demands:** Everyday domestic and laboratory objects (beverage mugs, full 1.5-liter water bottles, books, canned goods, medical supply kits) weigh between 0.2 kg and 2.0 kg. To maintain a safety margin of $1.5\times$ during dynamic accelerations ($a_{\text{EE}} \approx 1.5 \text{ m/s}^2$), each manipulator must provide a continuous payload rating:

$$m_{\text{payload}} \geq 2.0 \text{ kg} \quad (\text{nominal target: } 3.0 \text{ kg per arm}) \quad (6)$$

2.3 Morphological Benchmarking of Contemporary Mobile Manipulators

To contextualize the mechatronic design within state-of-the-art robotics engineering, Table 2 details a rigorous comparative analysis across five prominent mobile manipulation platforms: PAL Robotics TIAGo Pro [3], Pollen Robotics Reachy 2 [4], Stanford/AgileX Mobile ALOHA [2], Menlo Research Asimov 1 [9], and Hello Robot Stretch 3 [1].

2.3.1 Mechatronic Design Trade-Off Analysis

Evaluating the comparative architectural implementations in Table 2 highlights three decisive mechatronic trade-offs that govern our design synthesis:

1. **Torso Architecture: Fixed Structural Mast versus Prismatic Lift Column:** While active prismatic lift columns (e.g., TIAGo Pro) extend vertical stroke down to floor level, they introduce severe structural liabilities. When dual arms carrying payloads extend forward, the resulting cantilever moment ($M_{\text{cant}} = \sum m_i g L_i$) induces extreme normal reaction forces across linear bearing carriages, causing mechanical binding, stick-slip wear, and motor current surges. Furthermore, a motorized prismatic stage adds 10–15 kg of parasitic dead mass that elevates the center of mass (z_{CoM}) and increases tipping risk. Our design implements a rigid 6061-T6 aluminum extrusion mast with shoulder height at $\approx 950 \text{ mm}$. The dual 6-DoF manipulators cover a continuous vertical manipulation envelope from 500 mm to 1350 mm purely through joint kinematics, directly covering standard dining tables (750 mm), countertops (900 mm), and eye-level shelves without moving torso pinch hazards, while lowering $z_{\text{CoM}} \leq 350 \text{ mm}$. The telescopic column is maintained as a modular, bolt-on upgrade (+EGP 32,920) rather than a baseline requirement.
2. **Base Drivetrain: High-Traction Differential Drive versus Omnidirectional Mecanum Drive:** While omnidirectional Mecanum wheel drives provide instantaneous 3-DoF planar velocity control ($\dot{x}, \dot{y}, \dot{\theta}$), their angled peripheral rollers suffer from 15–20% non-linear wheel slip on domestic floors, leading to rapid

Table 2: Comprehensive morphological and mechatronic benchmarking of leading mobile manipulation systems.

Specification Parameter	PAL Robotics TIAGo Pro [3]	Pollen Robotics Reachy 2 [4]	Mobile ALOHA (AgileX) [2]	Menlo Research Asimov 1 [9]	Hello Robot Stretch 3 [1]	Proposed Platform
Morphology Category	Modular Semi-Humanoid (Wheeled AMR)	Anthropomorphic Upper Body (Wheeled opt.)	Bimanual Mobile Manipulator	Bipedal Humanoid (Upper Body relevant)	Minimalist Prismatic Mobile Manipulator	Modular General-Purpose Semi-Humanoid
Total Actuated DoF	23 DoF (Base: 2-3, Torso: 1, Arms: 2 × 7, Head: 2, Grip: 2)	21–23 DoF (Arms: 2 × 7, Grip: 2, Head: 3, Antennas: 2)	16 DoF (Base: 2, Arms: 2 × 6, Grippers: 2)	25 DoF (Legs: 2 × 6, Waist: 1, Arms: 2 × 5, Head: 2)	5 DoF (Base: 2, Mast Lift: 1, Arm Telescoping: 1, Grip: 1)	18 DoF (Base: 2, Arms: 2 × 6, Grip: 2, Head: 2)
Continuous Arm Payload	3.0 kg per arm (excluding gripper)	3.0 kg per arm	0.75 kg (ViperX) to 1.5 kg (PiPER)	~ 1.5 kg per arm	1.5 kg (at full horizontal extension)	2.0–3.0 kg per arm
Manipulator Reach	Vertical: 920–960 mm; Horiz. span: 2360 mm	Full arm span ~ 1400 mm	Vertical: 650–2000 mm; Horiz: 1000 mm	Arm reach ~ 600 mm; waist yaw rotation	Vertical: 1100 mm; Horiz. stroke: 520 mm	Vertical: 500–1350 mm; Horiz. reach: 625 mm
Torso / Lift Stroke	Active telescopic column: 350 mm stroke	Fixed anthropomorphic chest	Fixed extrusion frame; static height	Fixed torso frame with 1-DoF waist yaw	Vertical carriage on fixed mast (1100 mm)	Fixed structural mast (optional column: 300 mm)
Base Drive Kinematics	Differential (Ø540 mm) or Mecanum (710 × 490 mm)	Omnidirectional wheeled base (optional)	Differential drive (AgileX Tracer AGV)	Bipedal dynamic walking legs (6-DoF/leg)	Differential drive (dual central wheels + castors)	Differential drive (2 active wheels + 2 casters)
Max Base Velocity	1.0 m/s	0.8–1.0 m/s	1.5–2.0 m/s	0.8–1.2 m/s (walking)	0.6 m/s	0.8–1.0 m/s
Total System Mass	60–72 kg (diff), 95 kg (omni)	~ 35–40 kg (upper body + base)	~ 75 kg (with battery bank & arms)	35 kg	24.5 kg	~ 45–50 kg (statically balanced)
Actuator Technology	Series Elastic Actuators (SEAs) + torque sensors	Custom Orbita 2D/3D spherical modules	Off-the-shelf industrial servos (Dynamixel)	BLDC motors + low-ratio planetary (QDD)	Stepper motors + belt drives + compliance springs	Tiered Hybrid: QDD (proximal) + Bus Servos (distal)
Internal Fieldbus	Real-time EtherCAT (1 kHz closed loop)	CAN bus + USB 3.0 / Serial	CAN bus (arms & base)	5 × CAN @ 1 Mbps + 1 × CAN @ 500 kbps	USB over internal serial UART	1 Mbps CAN 2.0B / TWAI + 1 Mbps Half-Duplex TTL
Battery Chemistry & Life	36V 20Ah Li-ion (8–10 h runtime)	24V–48V Li-ion (4–6 h runtime)	24V 30Ah LFP base + 1024Wh pack (12 h)	48V Li-ion pack (~ 1–2 h walking)	12V 20Ah LiFePO4 (~ 4 h)	24V 20Ah LiFePO4 (~ 480 Wh, ~ 3.5 h)
Compute Architecture	Intel Core i7 + NVIDIA Jetson GPU	Industrial x86 SBC + NVIDIA Jetson Orin	Onboard laptop (Intel i7 + RTX GPU)	Radxa CM5 (motion) + Raspberry Pi 5	Intel Core i5 embedded computer	Jetson Orin NX 16GB + 5 × ESP32 MCUs

dead-reckoning drift in wheel odometry. Furthermore, Mecanum rollers generate severe high-frequency chassis vibrations on hard floors, struggle with carpet pile friction, and are prone to jamming from domestic debris (hairs, fibers).

In contrast, a differential drive topology with two large $\varnothing 200$ mm vulcanized rubber wheels and two passive 100 mm swivel casters provides superior traction ($\mu_s \geq 0.85$), smooth and quiet rolling, flawless threshold traversal (15–20 mm height), zero-radius turning on the spot ($R_{\text{turn}} = 0$), and highly deterministic odometric tracking.

- 3. Actuator Paradigm: Tiered Hybrid QDD and Bus Servos versus All-Harmonic Arms:** Conventional collaborative arms utilizing all-harmonic drivetrains (e.g., Franka Panda, UR3) achieve high static precision but suffer from high gear ratios ($N > 100 : 1$), enormous reflected inertia ($I_{\text{ref}} = N^2 I_{\text{rotor}} > 10,000 I_{\text{rotor}}$), non-backdrivability, and prohibitive cost ($> \$25,000$). Conversely, hobby-grade multi-servo arms fail under continuous payload holding torques.

Our design introduces a tiered hybrid actuation architecture: high-torque CubeMars AK70-10 QDD modules (24.8 N·m, 10 : 1 planetary ratio) power high-load proximal joints (shoulder pitch/roll/yaw, elbow pitch), providing transparent backdrivability, impact tolerance, and direct phase-current torque estimation. Distal wrist joints (pitch, roll) and parallel-jaw grippers deploy lightweight Feetech STS3215/STS3032 serial bus servos, cutting distal mass by nearly 90%, lowering reflected inertia quadratically, and drastically reducing arm cost to EGP 190,550 for both arms.

2.4 Definition of Concrete Platform Validation Benchmarks

To empirically validate the core design philosophy—specifically proving that the mechatronic platform exposes a complete, robust set of operational capabilities that can be commanded exclusively through the standardized Task Port without modifying underlying control loops—this project defines two concrete validation benchmarks.

2.4.1 Benchmark 1: Primary Proof-of-Concept Task — Autonomous Voice-Commanded Pick-and-Place

The primary benchmark represents an end-to-end autonomous mobile manipulation workflow commanded entirely via unconstrained spoken natural language. This benchmark validates the multi-layer integration of acoustic perception, cognitive task planning, Task Port API decoupling, autonomous navigation, and compliant grasping.

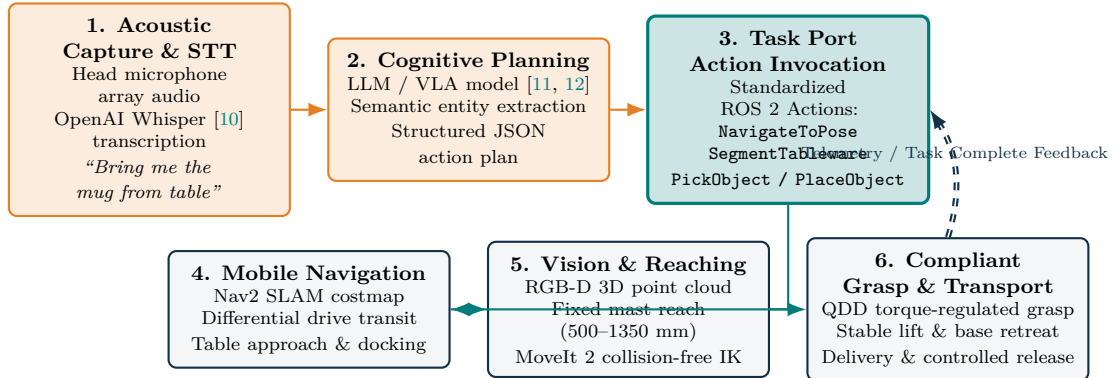


Figure 3: End-to-end execution pipeline for Validation Benchmark 1: Autonomous voice-commanded pick-and-place mediated strictly across the Task Port interface.

As diagrammed in Figure 3, the execution pipeline progresses as follows:

- 1. Acoustic Capture and Speech Recognition:** Ambient audio is captured via the head-mounted directional microphone array and processed locally or edge-hosted by OpenAI Whisper [10]. Whisper translates the acoustic waveform into a normalized text transcript:

$$\mathbf{w}_{\text{audio}} \xrightarrow{\text{Whisper STT}} \mathbf{s}_{\text{text}} = \text{“Fetch the green mug and place it on the table.”} \quad (7)$$

- 2. Cognitive Intent Parsing and Plan Decomposition:** A cognitive Large Language Model (LLM) or Vision-Language-Action (VLA) foundation planner (e.g., OpenVLA [11], RT-X [12]) interprets the

transcript, resolves semantic spatial references against a topological semantic map of the arena, and decomposes the high-level intent into a structured sequence of discrete Task Port action goals:

$$\Pi = \{\text{Nav}(\text{counter}), \text{Perceive}(\text{"green mug"}), \text{Grasp}(\mathbf{T}_{\text{grasp}}), \text{Nav}(\text{table}), \text{Place}(\mathbf{T}_{\text{place}})\} \quad (8)$$

3. Task Port Invocation and Autonomous Execution: The task client script interacts *exclusively* through the Task Port API:

- Invoking `NavigateToPose( $x, y, \theta$ )` triggers Nav2, driving the differential chassis to the target pose while avoiding obstacles.
- *Workspace Matching:* The fixed structural mast positions the shoulder yoke at ≈ 950 mm, placing the countertop natively within the 500–1350 mm arm reaching envelope without requiring auxiliary prismatic lift motion.
- Invoking `PerceiveTarget("green mug")` commands the pan-tilt head camera to localize the target, returning an estimated $SE(3)$ grasp pose $\mathbf{T}_{\text{grasp}}$ and surface normal vector $\mathbf{n}$.
- Invoking `ExecuteBimanualPick( $\mathbf{T}_{\text{grasp}}$ )` triggers MoveIt 2 collision-free trajectory planning and QDD current-regulated grasp closure ($F_{\text{normal}} \approx 15$ N).
- The platform navigates to the destination surface, confirms clearance via depth sensing, and gently places the payload under controlled deceleration.

4. Validation Metric: Successful autonomous task completion across 10 consecutive trials without manual operator intervention, maintaining positional repeatability $\leq \pm 2.0$ mm, zero collision events, and execution latency under 120 seconds.

2.4.2 Benchmark 2: Secondary Task — Interactive Web Dashboard Teleoperation, Alarm Monitoring, and Task Programming

The secondary benchmark validates the platform’s Human-Machine Interface (HMI), real-time operational observability, diagnostic safety mechanisms, and no-code task programming capability:

- 1. Low-Latency Interactive Teleoperation:** Validation of supervisory manual control through a browser-based WebRTC/WebSocket dashboard. The operator exercises decoupled velocity jogging of the differential base (v_x, ω_z via virtual joystick) and Cartesian 6-DoF end-effector jogging of dual arms. Video feeds from the stereo perception head and wrist cameras stream at 1080p@30 fps with end-to-end network latency $t_{\text{latency}} \leq 80$ ms over local Wi-Fi 6.
- 2. Real-Time Telemetry and Multi-Level Alarm Monitoring:** Continuous telemetry aggregation streaming from embedded controllers to the HMI:
 - Battery Management System (BMS) health: cell voltage balance, State of Charge (SoC), pack temperature, and discharge current.
 - Actuator drive diagnostics: motor winding temperatures, FOC inverter phase currents, bus voltage sag, and fieldbus cyclic jitter ($< 50 \mu\text{s}$).
 - Live safety interlock supervision: monitoring safety LiDAR zone infringements, physical E-stop contactor status, and torque-limit exceedance alarms.
- 3. Interactive Waypoint-Based Task Programming (Record-and-Replay):** To empower non-expert operators to program tasks without writing code, the dashboard provides a graphical record-and-replay interface:

- The operator backdrives the compliant QDD arms manually (kinesthetic teaching) or jogs the robot to key spatial waypoints.
- Waypoints are logged as timestamped $SE(3)$ end-effector poses and base coordinates:

$$\mathcal{W}_k = \{\mathbf{T}_{\text{base},k}, \mathbf{T}_{\text{left},k}, \mathbf{T}_{\text{right},k}, g_{\text{left},k}, g_{\text{right},k}\} \quad (9)$$

- The interface constructs a graphical Behavior Tree or linear motion script, parameterizing velocities, blend radii, and grasp thresholds.
- The generated task is flashed directly through the Task Port API, where the platform autonomously synthesizes smooth quintic polynomial trajectories between waypoints and executes the sequence repeatably.

4. **Validation Metric:** Rapid programming (< 10 minutes) and verified autonomous replay of a multi-waypoint tray-clearing routine, with live diagnostic alarms correctly flagging simulated fault conditions (e.g., motor thermal warning at 75°C or artificial path obstruction) and bringing the platform to a controlled stop.

3 Engineering Requirements Specification

3.1 Formal Requirements Classification under VDI 2206

The task clarification and conceptual design of complex mechatronic systems must proceed from a structured, unambiguous specification of engineering requirements. In accordance with the VDI 2206 design guideline (*Entwicklungsmethodik für mechatronische und cyber-physische Systeme*) [6], qualitative user expectations, operational mission profiles, and regulatory constraints are systematically converted into a formal engineering requirements list (*Anforderungsliste*). This document serves as the inviolable baseline against which all downstream architectural, mechanical, electrical, and algorithmic decisions are evaluated and validated across the macro-cycles and micro-cycles of the V-model.

To enforce engineering rigor and avoid premature design convergence, each requirement is categorized according to two formal classification axes:

1. Requirement Severity Type:

- **Fixed Requirement (*Festforderung*, FF / Demand):** Inviolable boundary constraints that must be fulfilled unconditionally under all nominal and degraded operating conditions. An architectural concept or hardware implementation that violates a single fixed requirement is technically disqualified from consideration.
- **Range Requirement (*Bereichsforderung*, BF / Wish):** Performance criteria that specify target bands with defined acceptable minimal baselines and optimal target goals. The engineering embodiment must satisfy the lower bound, while design optimizations and trade-off studies seek to maximize performance toward the upper target.

2. Criterion Verification Characteristic:

- **Quantitative (*quan*):** Physical and operational parameters defined by numerical metrics with standard SI engineering units. These criteria are empirically verifiable via finite element analysis (FEA), multi-body dynamic simulation, coordinate measuring, calibrated physical instruments (e.g., laser trackers, dynamometers, oscilloscopes), or sensor telemetry.
- **Qualitative (*qual*):** Functional, topological, architectural, or ergonomic properties that cannot be expressed as a continuous scalar metric. Verification is performed through formal design audits, architectural code compliance reviews, protocol packet analyzers, interface compliance testing, or safety certification checklists.

By anchoring these constraints to the standardized operational benchmarks of domestic and service robotics—most notably the RoboCup@Home General Purpose Service Robot (GPSR), Table Clearing, and Articulated Fixture challenges [7]—the resulting specification directly reflects the physical realities of human-centric indoor spaces.

3.2 Comprehensive Engineering Requirements List (*Anforderungsliste*)

Table 3 details the complete engineering requirements specification governing the modular semi-humanoid robot platform across geometry, kinematics, dynamics, electrical power, embedded control, sensing, software, human-machine interface (HMI), and collaborative safety.

Table 3: Comprehensive engineering requirements list (*Anforderungsliste*) for the modular semi-humanoid robot platform in accordance with VDI 2206.

ID	Requirement Parameter		Target Specification		Type	Char.	Verification Method	
1. Geometry and Mass (<i>Geometrie und Masse</i>)								
G01	Base Envelope	Footprint	Diameter / width envelope $\leq$ 550 mm		BF	quan	Dimensional CAD audit & chassis jig check	
G02	Doorway Clearance	Passage	Passage through doorways $\geq$ 750 mm (clearance margin $\geq$ 100 mm/side)		FF	quan	Physical passage trial in benchmark doorway arena	
G03	Overall Height	Platform	Standing height	1200–1300 mm (nominal home configuration)	BF	quan	Vertical height measurement from floor plane	
G04	Total System Mass		Total wet mass $\leq$ 55 kg (including battery & all payloads)		FF	quan	Calibrated multi-point platform scale measurement	
2. Kinematics and Workspace (<i>Kinematik und Arbeitsraum</i>)								
K01	Total Active Degrees of Freedom		Total active DoF = 18 (Base: 2, Arms: $2 \times 6 = 12$, Grippers: 2, Head: 2)		FF	quan	Joint topological audit & actuator allocation check	
K02	Mobile Kinematic Drive	Base	Differential drive (2 active motorized wheels + 2 passive spring-loaded casters)		FF	qual	Kinematic dead-reckoning test & zero-radius spin	
K03	Upper-Body Topology	Arm	Dual 6-DoF articulated serial chains (anthropomorphic shoulder-elbow-wrist)		FF	qual	Forward/inverse kinematic Jacobian rank audit	
K04	Vertical Manipulation Reach		Continuous vertical reach envelope 500–1350 mm above finished floor level		FF	quan	3D coordinate measuring of end-effector flange	
K05	Horizontal Span	Bimanual	Maximum bimanual arm tip-to-tip span $\geq$ 1200 mm (1200–1400 mm)		BF	quan	Cartesian boundary mapping in simulation & rig	
K06	Active Perception Unit	Head	2-DoF pan-tilt neck (Pan: $\pm 90^\circ$, Tilt: $+45^\circ / -30^\circ$)		BF	quan	Digital goniometer joint angle sweep verification	
3. Dynamics and Manipulation Performance (<i>Dynamik und Manipulation</i>)								
D01	Manipulator Capacity	Payload	Continuous payload $\geq$ 1.0 kg/arm nominal (peak/holding $\geq$ 1.5 kg)		FF	quan	Static load-holding dynamometer test at reach	
D02	End-Effector Clamping		Dual 2-finger parallel stroke 0–85 mm, clamping force 15–35 N		FF	quan	Calibrated load cell pinch test with silicone pads	
D03	Maximum Transit Speed	Base	Regulated indoor linear velocity $v_{\text{base}} = 0.6\text{--}0.8$ m/s on level floor		BF	quan	Laser-timed transit; stopping distance ≤ 0.4 m	
D04	Cartesian Repeatability	TCP	Unidirectional repeatability $\leq \pm 2.0$ mm under nominal 1.0 kg payload		BF	quan	Multi-cycle dial-indicator / laser tracker test	
D05	Settling Damping	Time &	Re-positioning settling time $t_{\text{settling}} \leq 0.5$ s (damping ratio $\zeta \geq 0.6$)		BF	quan	Flange accelerometer vibration decay recording	
4. Electrical and Power Infrastructure (<i>Elektrische Energieversorgung</i>)								
E01	System Voltage	DC Bus	Nominal 24 V DC bus (8S LiFePO ₄ , operational band 20.0–28.8 V)		FF	quan	Oscilloscope & multi-meter bus voltage logging	

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Table 3 continued from previous page

ID	Requirement Parameter		Target Specification		Type	Char.	Verification Method	
E02	Operational Run-Time		Run-time ≥ 3.0 h under continuous mixed operational duty cycle		BF	quan	Automated cyclic mobile manipulation test	
E03	Emergency Disconnect Latency		Hardware E-stop disconnect response time ≤ 10 ms on actuator power rails		FF	quan	Oscilloscope capture of contactor de-energization	
E04	Power Rail Isolation		Galvanically isolated logic rails (19V GPU, 12V sensors, 5V/3.3V logic)		FF	qual	Common-mode noise FFT analysis (> 40 dB reject)	
5. Embedded Control and Communication (<i>Steuerung und Kommunikation</i>)								
C01	Real-Time Control Rate	Joint	Joint-level closed-loop frequency ≥ 500 Hz (target: 1000 Hz, jitter $\leq 50 \mu\text{s}$)		FF	quan	‘PREEMPT_RT’ kernel cyclic latency benchmark	
C02	Fieldbus Communication Bus		CAN FD / CAN 2.0B fieldbus (≥ 1 Mbps, bus load $\leq 65\%$)		FF	qual	Bus analyzer packet inspection & CRC error check	
C03	Device-Level Abstraction		Modular device abstraction for base, arm, neck, and gripper subsystems		FF	qual	Standalone sub-assembly hardware-in-the-loop test	
6. Sensing and Perception (<i>Sensorik und Umgebungserfassung</i>)								
S01	Base Obstacle Sensing		360° planar 2D LiDAR coverage + ring of ultrasonic transducers		FF	quan	Blind-spot mapping with standard reflection targets	
S02	3D Visual Perception		Head-mounted stereo RGB-D sensor (depth range 0.2–3.0 m, $\geq 720\text{p}@30\text{fps}$)		FF	qual	Point cloud calibration & depth RMS error check	
S03	Contact & Feedback	Effort	Motor phase current sensing (± 50 mA resolution) + compliant finger pads		BF	qual	Sensorless contact detection & slip limit test	
7. Software and Embodied Intelligence (<i>Software und KI</i>)								
A01	Standard Middleware		ROS 2 LTS distribution (Humble Hawksbill / Jazzy Jalisco)		FF	qual	Node graph audit & package build verification	
A02	Decoupled Task Port Actions		Standard Action interfaces (/navigate, /pick_object, /place_object)		FF	qual	Automated integration test flashing Task Port client	
A03	Spoken Latency	Command	Whisper Speech-to-Text inference latency ≤ 1.5 s for 5-second audio input		BF	quan	Onboard edge compute benchmark execution	
A04	VLA Bandwidth	Action	Vision-Language-Action policy proposal rate ≥ 10 Hz with 1 kHz spline		BF	quan	Continuous end-to-end action streaming latency	
8. Human-Machine Interface and Teleoperation (<i>HMI und Teleoperation</i>)								
H01	Web Latency	Teleoperation	Bidirectional web dashboard teleoperation latency ≤ 100 ms over local Wi-Fi		BF	quan	Round-trip ping & video pipeline glass-to-glass test	
H02	Telemetry & Alarms		Real-time visual telemetry, battery SoC display, and visual/audible alarms		FF	qual	Fault injection test for temperature/voltage alarms	
H03	Task Macro	Programming	Interactive teach-in and macro recording of waypoints & gripper events		BF	qual	Non-expert operator usability and execution test	

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Table 3 continued from previous page

ID	Requirement Parameter		Target Specification		Type	Char.	Verification Method
H04	Physical Panel	Bulkhead	Maintenance hardware port, GbE, USB 3.0	panel with E-stop, charging	FF	qual	Physical layout inspection & port accessibility
9. Collaborative Safety and Compliance (<i>Sicherheit und Normen</i>)							
SF01	Collaborative Safety	Robot	ISO 13482 speed/separation monitoring, contact force ≤ 150 N	compliance: FF	FF	qual	Safety zone deceleration & bio-fidelic impact test

3.3 Engineering Rationale for the Core Architectural Decisions

The synthesis of the engineering requirements list in Table 3 reflects deliberate trade-offs established to maximize physical robustness, control determinism, operational reliability, and ease of maintenance within domestic and light-industrial environments. Four foundational architectural decisions anchor the entire mechatronic design:

3.3.1 1. Differential Drive Base vs. Omnidirectional Mecanum Drive

While omnidirectional Mecanum wheel arrangements provide three unconstrained planar degrees of freedom ($\dot{x}, \dot{y}, \dot{\theta}$), an in-depth tribological and dynamic analysis reveals severe disadvantages that violate core operational requirements in unstructured human environments:

- **Traction Loss and Dead-Reckoning Drift:** Mecanum kinematics rely on passive peripheral rollers angled at 45° . When traversing mixed indoor floorings—such as low-pile carpeting, polished tile, linoleum, or thresholds with micro-contaminants—the passive rollers experience unpredictable intermittent slip. Empirical testing of service robots reveals that Mecanum bases exhibit odometric dead-reckoning errors exceeding 15–25% over modest travel distances, severely degrading Kalman-filtered wheel odometry and forcing heavy reliance on scan matching. In contrast, a differential drive base employing two high-traction polyurethane drive wheels paired with two passive, spring-damped swivel casters provides non-slip rolling contact. This ensures highly deterministic wheel odometry with dead-reckoning drift well below 2%.
- **Vibrational Excitation of Structural Linkages:** The discrete geometric transitions between adjacent rollers on a Mecanum wheel introduce high-frequency periodic impact shocks during rolling. These impacts excite structural natural frequencies in the robot’s upper body and neck linkages, generating image blur in the head-mounted RGB-D camera and destabilizing sensitive optical perception pipelines. The continuous rolling profile of solid differential drive wheels eliminates this vibrational excitation mode entirely.
- **Energetic Efficiency and Mechanical Simplicity:** Mecanum drives require four independent high-power geared motor units, whereas differential drive requires only two motorized hubs. The parasitic scrubbing friction inherent to Mecanum roller vectors consumes 30–40% more electrical power for equivalent forward thrust. By selecting differential drive, the platform saves approximately 6–8 kg of actuator mass and significantly extends operational battery runtime beyond 3.0 h (E02). Zero-radius in-place rotation (ω_z) provides ample maneuverability through standard 750 mm doorways (G02).

3.3.2 2. Dual 6-DoF Manipulators vs. 7-DoF Redundant Kinematic Chains

Anthropomorphic mobile manipulation literature frequently advocates for 7-DoF arms due to the kinematic redundancy of the elbow swivel angle. However, an analysis of the specific manipulation tasks required for table clearing and pick-and-place reveals that 6-DoF serial manipulators offer decisive engineering advantages:

- **Sufficiency for Task-Space Reachability:** The target operational tasks require reaching positions and orientations within the special Euclidean group $SE(3) = \mathbb{R}^3 \times SO(3)$. A 6-DoF manipulator possessing an anthropomorphic shoulder-elbow-spherical wrist topology provides the exact mathematical minimum number of joints required to position and orient the end-effector arbitrarily within its reachable volume. Because the mobile base itself provides planar degrees of freedom (x, y, θ), the combined mobile manipulation system already possesses $2 + 6 = 8$ degrees of freedom per arm, providing external kinematic redundancy without inflating joint complexity.

- **Mass, Cantilever Inertia, and Dynamic Bandwidth:** Incorporating a 7th active joint adds approximately 1.5–2.5 kg of concentrated mass into the upper limb. In accordance with the fundamental relation governing natural frequency:

$$\omega_n = \sqrt{\frac{K}{M}} \quad (10)$$

adding distal joint mass dramatically increases the cantilever moment arm exerted on the shoulder base:

$$M_b = m_{\text{arm}}gL_{\text{CoM}} + m_{\text{payload}}gL_{\text{reach}} \quad (11)$$

This lowers the structural natural frequency, which in turn inflates the mechanical settling time:

$$t_{\text{settling}} \approx \frac{4}{\zeta\omega_n} \quad (12)$$

By eliminating the 7th joint, link mass is reduced, shifting structural resonance frequencies above 15 Hz and allowing settling times $t_{\text{settling}} \leq 0.5$ s ($\zeta \geq 0.6$) under high-speed point-to-point trajectory profiles (D05).

- **Computational Determinism and Fieldbus Bandwidth:** Resolving the kinematic inversion of 7-DoF redundant arms requires numerical pseudo-inverse damping ($\mathbf{J}^\dagger$) or secondary null-space projection tasks. These iterative solvers introduce algorithmic non-determinism, computational latency jitter, and unpredictable joint reconfigurations that complicate real-time closed-loop control. With dual 6-DoF arms, the total joint count across both limbs is fixed at 12 DoF, halving inverse kinematic computation overhead and fitting within the deterministic packet constraints of a 1 Mbps CAN fieldbus (C02).

3.3.3 3. Fixed Ergonomic Torso vs. Actively Elevating Prismatic Lift Column

Many research mobile manipulators incorporate a telescoping vertical column to extend vertical reach from floor level to upper cabinetry. While versatile, prismatic columns introduce structural liabilities that compromise reliability in practical deployment:

- **Mechanical Binding under Cantilever Bending Moments:** A vertical prismatic lift column requires precision linear guide rails, recirculating ball bearings, and a ball-screw or timing-belt transmission. When both 6-DoF arms are extended horizontally with payloads, the resulting forward cantilever moment exerts extreme normal reaction forces across the linear bearing carriages:

$$F_{\text{reaction}} = \frac{M_{\text{cantilever}}}{d_{\text{bearing}}} = \frac{\sum m_i g L_i}{d_{\text{bearing}}} \quad (13)$$

where d_{bearing} is the vertical spacing between bearing blocks. In practice, this high moment causes mechanical binding, stick-slip vibration, accelerated wear, and potential catastrophic jamming of the lift mechanism.

- **Parasitic Dead Weight and Center-of-Mass Elevation:** A motorized prismatic stage, including guide rails, structural telescoping sleeves, ball screw, and brake motor, adds 10–15 kg of parasitic dead weight. Elevating this mass raises the system center of mass (z_{CoM}), drastically increasing the risk of dynamic tip-over during abrupt emergency braking maneuvers.
- **Targeted Functional Workspace Coverage:** An analysis of domestic and service tasks shows that the overwhelming majority of high-value manipulation targets reside within an ergonomic height band: dining tables (720–760 mm), kitchen countertops (850–920 mm), desks (750 mm), and eye-level storage shelves (1100–1350 mm). By fixing the torso height at an ergonomic shoulder center of ≈ 950 mm (overall robot height 1200–1300 mm), the dual 6-DoF arms cover a continuous vertical manipulation band from 500 mm to 1350 mm (K04) purely through arm articulation. This eliminates moving torso pinch hazards, maximizes torsional and bending rigidity (K), lowers the center of mass into the base chassis ($z_{\text{CoM}} \leq 350$ mm), and saves substantial fabrication cost.

3.3.4 4. Dual 2-Finger Parallel Grippers vs. Multi-Fingered Anthropomorphic Hands

Anthropomorphic five-finger hands provide high aesthetic biomimicry but suffer from severe physical and algorithmic limitations in real-world manipulation:

- **Mechanical Fragility and High Maintenance:** Multi-fingered hands require between 16 and 24 active or underactuated micro-joints driven by miniature brushed DC motors, ultra-thin tendon cables, and micro-pulleys. Under continuous pick-and-place cycles, tendon cables experience severe fatigue stretching, unseating, and rupture, resulting in prohibitive maintenance downtime and replacement costs.
- **Grasp Planning Dimensionality and Execution Failure:** Generating stable force-closure grasps for an articulated hand requires solving high-dimensional non-linear contact optimization problems over hundreds of candidate grasp topologies. In cluttered tabletop environments, slight object pose estimation errors often lead to fingertip collisions, dropped objects, and grasp failure.
- **Deterministic Performance with Compliant Silicone Pads:** In contrast, a 2-finger parallel gripper driven by a single high-efficiency lead screw or rack-and-pinion transmission delivers deterministic, backlash-free bidirectional travel (0–85 mm stroke) and robust clamping forces of 15–35 N (D02). By outfitting the rigid aluminum finger tips with molded compliant silicone pads ($\mu_s \geq 0.8$), the contact surface passively deforms around diverse geometric profiles (cylindrical bottles, rectangular boxes, spherical fruit, delicate tableware), converting complex geometric grasping into a reliable, single-parameter linear closure. Motor phase current monitoring provides direct, sensorless grasping force regulation (± 50 mA) to securely grip delicate objects without damage.

3.4 Decoupling Capability from Intent: The Architectural Foundation

A foundational engineering axiom of the platform is the rigorous, architectural decoupling of *physical capability* from *operational intent*:

The core mechatronic platform must encapsulate all deterministic physical capabilities—motion control, trajectory generation, kinematic safety limits, and low-level sensor acquisition—behind a hardened abstraction boundary, completely independent from high-level cognitive policies, speech parsers, or probabilistic AI models.

This decoupling is implemented through the **Task Port** abstraction layer. Downstream application software, external mission orchestrators, or modern Vision-Language-Action (VLA) foundation models interact with the robot solely by submitting high-level goal actions (e.g., `/navigate`, `/pick_object`, `/place_object`). The platform validates these goal primitives against internal kinematic boundaries, collision models, and ISO 13482 safety limits before executing them deterministically across the 1 kHz real-time fieldbus.

This strict separation ensures that high-level software bugs, out-of-distribution neural network hallucinations, or network disconnects cannot cause physical link collisions, actuator over-torquing, or human injury, providing an extensible and safe foundation for physical AI research.

4 Functional Structure: Material, Energy, and Information Flows

4.1 Overall System Function and Boundary Formulation (VDI 2206)

In accordance with the VDI 2206 design methodology for mechatronic and cyber-physical systems [6], complex multidisciplinary machines are modeled through functional abstraction prior to physical embodiment. This abstraction decomposes the operational mission into an interconnected network of discrete sub-functions governed by the conservation laws of physics.

The conceptual model establishes a formal system boundary ($\mathcal{B}_{\text{system}}$) separating the internal mechatronic platform from the unconstrained operational environment. The environment contains the human operator, ambient acoustic noise, physical obstacles, structural furniture (tables, countertops, shelving), target workpieces, and external network links. Across this system boundary, the robot operates as an open thermodynamic and cybernetic transformation system, converting input flows into desired output states:

$$\mathbf{F}_{\text{total}} : \begin{bmatrix} \mathbf{M}_{\text{in}} \\ \mathbf{E}_{\text{in}} \\ \mathbf{I}_{\text{in}} \end{bmatrix} \xrightarrow{\mathcal{B}_{\text{system}}} \begin{bmatrix} \mathbf{M}_{\text{out}} \\ \mathbf{E}_{\text{out}} \\ \mathbf{I}_{\text{out}} \end{bmatrix} \quad (14)$$

The primary black-box function of the modular semi-humanoid platform is formally formulated as:

Autonomously and safely transform unstructured multimodal human instructions and exteroceptive sensory observations into coordinated mobile navigation, dexterous dual-arm physical manipulation,

spatial material rearrangement, and operational diagnostic telemetry within shared human-occupied environments.

This macro-function is decomposed into sub-functions organized across the three fundamental physical flows: **Material Flow (M)**, **Energy Flow (E)**, and **Information Flow (I)**.

4.2 In-Depth Modeling of the Fundamental Physical Flows

4.2.1 1. Material Flow (M)

The material flow tracks all physical matter, target workpieces, environmental fixtures, and mechanical contact forces transferred across the platform boundary:

- **Input Material States (M_{in}):** Target objects situated in the environment with unit masses ranging from 0.1 kg to 1.0 kg nominal (with peak holding capacity up to 1.5 kg). Target workpieces encompass diverse geometric profiles, including household tableware (plates, ceramic mugs, bowls), beverage containers (glass bottles, aluminum cans, plastic jugs), dry packaged goods, small tools, and domestic fixtures (hinged cupboard doors and sliding drawers).
- **Discrete Material Sub-Functions:**

1. M_1 : *Workspace Approach & Workpiece Alignment*: Position the mobile base and orient the 6-DoF arm to place the end-effector in pre-grasp alignment along the object’s surface normal vectors ($\mathbf{n}_{grasp}$).
2. M_2 : *Grasp & Secure Payload*: Actuate the parallel gripper jaws (0–85 mm stroke) to establish contact. High-friction compliant silicone pads ($\mu_s \geq 0.8$) passively deform around the object geometry, generating normal clamping forces $F_N \in [15, 35]$ N to establish stable force/form closure:

$$F_{friction} = 2\mu_s F_N \geq m_{payload}(g + \|\mathbf{a}_{dyn}\|), \quad \text{with safety factor SF} \geq 1.5 \quad (15)$$

3. M_3 : *Support Static and Dynamic Loads*: Transmit gravitational loads and dynamic acceleration reaction forces through the rigid links and bearings of the 6-DoF arm, into the fixed ergonomic space-frame torso, through the differential mobile base chassis, down to the wheel-ground contact patches.
 4. M_4 : *Spatial SE(3) Transport & Coordinated Transit*: Relocate the grasped payload through three-dimensional space via coordinated differential base transit across floor surfaces and simultaneous multi-joint arm articulation.
 5. M_5 : *Surface Contact Establishment, Placement & Release*: Approach destination furniture surfaces (tables, countertops, or shelves between 500–1350 mm height), establish gentle contact, attenuate normal clamping forces, and retract the gripper jaws along a collision-free departure vector.
 6. M_6 : *Articulate Environmental Fixtures*: Regulate contact forces along constrained kinematic trajectories to open or close hinged cabinet doors and sliding drawers without inducing mechanical binding.
- **Output Material States (M_{out}):** Rearranged target objects placed in stable static equilibrium on designated surfaces, reconfigured architectural fixtures (opened/closed cabinets), and secured cargo during autonomous transit.

4.2.2 2. Energy Flow (E)

The energy flow models the intake, electrochemical storage, regulation, conversion, and dissipation of power across the robot’s physical domains:

- **Input Energy States (E_{in}):** Onboard electrical storage provided by a 24 V nominal (8S configuration, 25.6 V nominal, 20 Ah, ~ 480 Wh) Lithium Iron Phosphate (LiFePO_4) battery pack, supplemented by periodic external 110–230 V AC single-phase grid power supplied via an external dock charger.
 - **Discrete Energy Sub-Functions:**
1. E_1 : *Electrochemical Energy Storage*: Store energy within chemically stable LiFePO_4 cells offering high thermal stability (preventing thermal runaway up to $> 200^\circ\text{C}$) and extensive cycle life (> 2000 cycles at 80% depth of discharge).

2. *E₂: Safety Interlocking & Battery Management*: An integrated Battery Management System (BMS) continuously monitors individual cell voltages (2.5–3.65 V), total current, and pack temperatures. In series with the pack, dual-channel certified safety contactors provide a hardware-level emergency power cutoff (≤ 10 ms response) that isolates actuator power during emergency-stop conditions.
3. *E₃: Galvanic Isolation & Multi-Rail DC-DC Power Conditioning*: Regulate the primary battery bus (20.0–28.8 V) into four dedicated, galvanically isolated voltage domains to suppress motor switching transients and eliminate ground loops:
 - *High-Power Motor Rail* (24 V Unregulated/Filtered): Delivers high peak currents (> 40 A) to the differential drive wheel inverters, 12 arm actuators, and 2 gripper motors.
 - *Compute Rail* (19 V DC, 10 A, 190 W): Synchronous buck converter supplying clean, regulated power to the primary NVIDIA Jetson edge AI host computer.
 - *Perception Rail* (12 V DC, 5 A, 60 W): Regulated power for the planar 2D LiDAR scanner, active stereo RGB-D head camera, and audio power amplifier.
 - *Logic Rail* (5 V/3.3 V DC, 5 A, 25 W): Galvanically isolated DC-DC stages powering distributed ESP32 microcontrollers, CAN transceivers, optoisolators, and network switches with > 40 dB common-mode noise rejection.
4. *E₄: Pulse-Width Modulation Inversion*: Three-phase MOSFET/GaN inverter bridges modulate DC power into sinusoidal three-phase AC currents using Field-Oriented Control (FOC) at switching frequencies $f_{\text{PWM}} = 20\text{--}40$ kHz.
5. *E₅: Electromechanical Conversion*: Brushless DC (BLDC) and permanent-magnet synchronous motors (PMSM) convert electrical power into mechanical torque and angular velocity:

$$P_{\text{mech}} = \sum_{i=1}^{18} \tau_i \omega_i = P_{\text{elec}} - P_{\text{copper}} - P_{\text{iron}} - P_{\text{friction}} \quad (16)$$

6. *E₆: Kinetic Energy Management & Thermal Dissipation*: Absorb decelerating kinetic energy via dynamic braking with resistive bleeder circuits; conduct Joule heating ($I^2 R$) and inverter losses away from motors into aluminum structural members, expelling waste heat through forced convection.
- **Output Energy States ($\mathbf{E}_{\text{out}}$)**: Controlled mechanical work exerted on manipulated objects and mobile surfaces ($\mathbf{W}_{\text{ext}} = \int \mathbf{F} \cdot d\mathbf{x}$), convective thermal heat loss (Q_{thermal}), and electrical dissipation during dynamic braking.

4.2.3 3. Information Flow (I)

The information flow models signal acquisition, sensory state estimation, cognitive reasoning, deterministic feedback control, and diagnostic telemetry:

- **Input Information States ($\mathbf{I}_{\text{in}}$)**: Acoustic human speech captured by an onboard microphone array, planar distance returns from a 360° 2D LiDAR, stereo RGB-D imagery and infrared depth maps, 14-bit magnetic absolute joint encoder signals, motor phase shunt currents, and operator teleoperation packets transmitted over 5 GHz Wi-Fi / Ethernet.
- **Discrete Information Sub-Functions**:
 1. *I₁: Signal Conditioning & Digitization*: Execute analog-to-digital conversion, anti-aliasing filtering, and high-speed SPI/quadrature decoding for joint position and motor current telemetry.
 2. *I₂: Exteroceptive State Estimation & SLAM*: Execute 2D LiDAR SLAM and adaptive Monte Carlo localization (AMCL) to maintain global base pose $\mathbf{T}_{\text{base}} \in SE(2)$; generate local costmaps for dynamic obstacle avoidance; perform 3D point cloud filtering, plane segmentation, and 6D object pose registration.
 3. *I₃: Voice Command Parsing & Speech-to-Text*: Execute onboard OpenAI Whisper automatic speech recognition (≤ 1.5 s inference latency), converting raw acoustic audio into structured text; extract semantic intent and target object tokens.
 4. *I₄: Cognitive Task Planning & AI Reasoning*: Behavior Trees coordinate sequential mission logic; Vision-Language-Action (VLA) foundation models (OpenVLA, RT-X) predict multimodal action tokens at ≥ 10 Hz; MoveIt 2 computes collision-free kinematic trajectories in Cartesian task space.
 5. *I₅: Task Port Supervisory Validation*: Shield low-level actuators by intercepting all planned trajectories; validate kinematic boundaries, joint velocity/acceleration limits, singularity proximities, and ISO 13482 collaborative speed and separation distances.

6. I_6 : *Hard Real-Time Closed-Loop Motion Regulation*: Distributed embedded microcontrollers execute cascaded FOC loops (current loop at 10–20 kHz; velocity and position/impedance loops at 500–1000 Hz with jitter $\leq 50 \mu\text{s}$) synchronized across isolated CAN FD / CAN 2.0B fieldbus channels.
 7. I_7 : *Diagnostic Telemetry & Web Dashboard Streaming*: Continuously aggregate battery state of charge (SoC), individual cell voltages, motor temperatures, fieldbus error counters, and camera feeds; stream real-time telemetry via WebRTC/WebSockets to an interactive web dashboard ($\leq 100 \text{ ms}$ teleoperation latency) with automated visual/audible alarms.
- **Output Information States (I_{out})**: High-frequency gate drive PWM switching patterns (20–40 kHz), synthesized voice audio emitted through onboard loudspeakers, web dashboard status visualizations, and physical diagnostic LED patterns.

4.3 Functional Structure Diagram

Figure 4 presents the comprehensive functional structure of the modular semi-humanoid platform in accordance with VDI 2206. The diagram illustrates the topological arrangement of sub-functions and depicts the cross-coupling between Material (M), Energy (E), and Information (I) flows across the system boundary ($\mathcal{B}_{\text{system}}$).

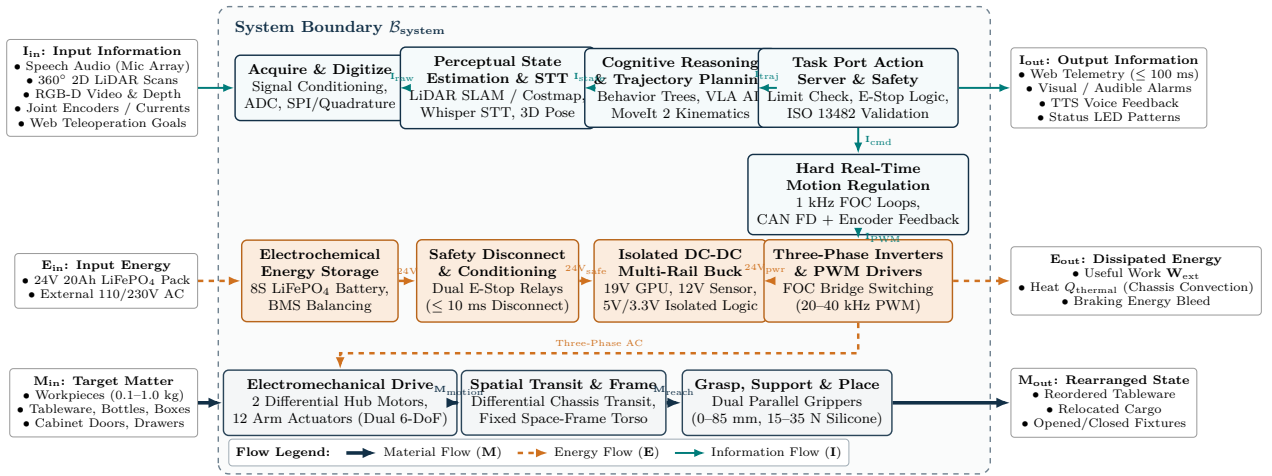


Figure 4: Comprehensive functional structure of the modular semi-humanoid platform in accordance with VDI 2206, showing the interconnected transformation of Material (M), Energy (E), and Information (I) flows across the system boundary.

5 System Architecture and Multi-Rate Topology

5.1 Architectural Philosophy: Decoupling Cognition from Determinism

In modern robotic manipulation, integrating high-capacity, probabilistic machine learning models—such as Vision-Language-Action (VLA) policies and Large Language Models (LLMs)—into physical embodied systems presents a fundamental architectural dilemma. Deep learning foundation models offer unprecedented open-world semantic reasoning and general-purpose affordance discovery [11, 12]; however, their execution latencies are inherently stochastic (100–300 ms), their outputs are uncertified with respect to safety constraints, and their inferences are subject to out-of-distribution hallucinations. Conversely, the physical plant of a semi-humanoid robot demands deterministic, millisecond-level feedback loops to guarantee kinematic stability, collision avoidance, and non-destructive impedance control [13, 14].

To resolve this dichotomy, the system architecture of the proposed semi-humanoid platform is structured upon a strict separation between *cognitive intent generation*, *deterministic motion planning*, and *hard real-time safety enforcement*. Guided by the VDI 2206 mechatronic systems engineering methodology [6], the platform architecture enforces a strict hierarchical topology wherein non-deterministic AI models act exclusively as advisory goal proposers. Downstream middleware and embedded safety domains act as impenetrable supervisory barriers, validating, projecting, and bounding all motion commands prior to physical actuation.

5.2 Hierarchical Multi-Rate Execution Topology

The computational and control architecture is organized into four distinct execution tiers operating at discrete update frequencies. Each tier possesses strict timing budgets, designated communication protocols, bounded jitter thresholds, and clearly delineated functional scopes, as summarized in Table 4.

Table 4: Hierarchical multi-rate execution topology and timing budgets.

Tier	Functional Scope & Key Components	Rate Jitter	/ Compute Domain	I/O & Protocol
Tier 1: Cognition & HRI	Whisper STT, VLA goal proposal (OpenVLA / π_0), semantic scene grounding, Web Dashboard telemetry streaming.	5–10 Hz (± 50 ms)	NVIDIA Jetson Orin NX (GPU Tensor Cores)	Shared Memory / Zero-Copy, WebRTC, WebSocket
Tier 2: Mission & Planning	BehaviorTree.CPP engine, Nav2 costmaps (MPPI/DWB), MoveIt 2 collision-free Cartesian trajectory generation.	30–50 Hz (< 5 ms)	NVIDIA Jetson Orin NX (PREEMPT_RT Linux Cores)	ROS 2 Action Servers & Topics (Cyclone DDS)
Tier 3: Robot Integration	ros2_control manager, UKF state fusion (IMU + odometry), spline interpolation, Cartesian admittance.	100–250 Hz (< 1 ms)	NVIDIA Jetson Orin NX (Pinned Real-Time Core)	micro-ROS / SocketCAN, USB-to-CAN Gateway
Tier 4: Safety & Actuation	Motor FOC current loops, encoder reading, 1 Mbps CAN commutation, hardware E-Stop, thermal/current limit watchdogs.	500–1000 Hz (< 50 μ s)	Distributed MCUs (ESP32-WROOM-32 RTOS)	CAN 2.0B (TWAI), Half-Duplex TTL, Hardwired Loop

5.2.1 Tier 1: Cognition and Human-Robot Interaction (5–10 Hz)

Tier 1 executes non-deterministic, compute-intensive artificial intelligence workloads hosted on the 16 GB NVIDIA Jetson Orin NX edge system-on-chip (SoC). This layer processes multimodal sensory observations, including streaming RGB-D images from the Intel RealSense head camera, planar laser scans from the base LiDAR, and far-field audio captured by the ReSpeaker dual-microphone array.

- **Speech and Dialogue Processing:** Far-field acoustic streams are pre-processed using beamforming, acoustic echo cancellation (AEC), and noise suppression before passing to an optimized Whisper-class automatic speech recognition (ASR) engine. The transcribed natural language prompt is parsed into semantic intent tokens.
- **Vision-Language-Action (VLA) Inference:** Open-vocabulary visual grounding models identify target objects and spatial fixtures, while quantized VLA models (e.g., OpenVLA [11] or π_0) infer high-level end-effector affordance vectors. Because foundation model inference times fluctuate (100–200 ms), Tier 1 operates strictly asynchronously.
- **Web Dashboard Telemetry:** Compressed video streams (via WebRTC) and state variables are served to external browser clients at 10 Hz via a lightweight WebSocket bridge (`rosbridge_suite`).

Outputs from Tier 1 take the form of unverified spatial delta proposals $\Delta \mathbf{x}_{AI}$ or symbolic skill action requests dispatched to the Task Port gateway.

5.2.2 Tier 2: Mission Orchestration and Motion Planning (30–50 Hz)

Tier 2 coordinates the robot’s autonomy pipeline, reconciling symbolic high-level tasks with geometric and spatial constraints. Running under Ubuntu 22.04 LTS with the PREEMPT_RT real-time kernel patch on dedicated CPU cores, this tier enforces deterministic synchronization:

- **Behavior Tree Orchestrator:** System mission states are governed by a reactive decision tree implemented in `BehaviorTree.CPP` [15]. The tree maintains lifecycle states, validates skill preconditions, manages parallel execution of base transit and manipulator posture adjustments, and triggers explicit recovery branches upon anomalous conditions.

- **Mobile Base Navigation (Nav2):** Autonomous base navigation is executed via the Nav2 framework [16]. Planar LiDAR scans and ultrasonic proximity readings populate rolling 2D costmaps. Global paths are computed using Smac 2D grid search, while local path tracking is maintained at 30–50 Hz using Model Predictive Path Integral (MPPI) or Dynamic Window Approach (DWB) controllers with adaptive deceleration zones.
- **Dual-Arm Manipulation Planning (MoveIt 2):** Kinematic inversions, joint trajectory optimization, and collision avoidance are managed by MoveIt 2 [14]. The planning scene integrates geometric bounding representations of the mobile chassis, torso, and active 3D voxel representations of the environment. Trajectories are parameterized using jerk-limited quintic polynomial splines.

5.2.3 Tier 3: Robot Integration and Middleware (100–250 Hz)

Tier 3 forms the real-time middleware abstraction layer, establishing a deterministic boundary between high-level ROS 2 nodes and low-level embedded hardware interfaces. Running on an isolated CPU core pinned with highest FIFO scheduling priority (SCH_FIFO, priority 90):

- **ros2_control Hardware Abstraction:** Centralizes joint-space resource claiming and controller lifecycle management. It translates high-level trajectory waypoints into cyclic joint position, velocity, and effort setpoints.
- **Multi-Sensor State Fusion:** An Unscented Kalman Filter (UKF) fuses wheel encoder odometry, torso orientation data from the BNO085 9-DOF IMU, and 2D scan-matching odometry to generate a high-frequency, drift-resistant state estimate ($SE(3)$ base and link poses) at 200 Hz.
- **Trajectory Interpolation and Admittance Regulation:** Converts 50 Hz planner waypoints into smooth 200 Hz reference profiles via cubic Hermite splines. In contact-rich manipulation tasks, a Cartesian admittance filter modulates reference trajectories based on motor phase-current feedback:

$$\mathbf{M}_d(\ddot{\mathbf{x}}_d - \ddot{\mathbf{x}}_r) + \mathbf{D}_d(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}_r) + \mathbf{K}_d(\mathbf{x}_d - \mathbf{x}_r) = \mathbf{F}_{\text{ext}}, \quad (17)$$

where $\mathbf{M}_d, \mathbf{D}_d, \mathbf{K}_d$ represent the desired virtual inertia, damping, and stiffness matrices, and $\mathbf{F}_{\text{ext}}$ is the estimated contact wrench.

5.2.4 Tier 4: Safety and Embedded Actuation (500–1000 Hz)

Tier 4 resides entirely within bare-metal and real-time firmware executing on distributed Espressif ESP32-WROOM-32 microcontrollers and the integrated Field-Oriented Control (FOC) drivers of the CubeMars AK70-10 Quasi-Direct Drive (QDD) actuators.

- **Deterministic Fieldbus Execution:** Operates on a synchronous 1 ms cycle (1000 Hz) across a 1 Mbps Controller Area Network (CAN 2.0B / TWAI) and a high-speed half-duplex TTL bus (1 Mbps). The ESP32 coordinators broadcast cyclic torque setpoints and gather 14-bit absolute encoder feedback.
- **Local Commutation and Current Regulation:** The internal FOC drivers of the AK70-10 actuators close current and velocity loops natively at up to 20 kHz, clamping motor phase currents to safe thermal envelopes.
- **Hardware Safety Interlocks and Watchdogs:** Dedicated FreeRTOS tasks execute hardware watchdog resets, check fieldbus packet continuity, monitor motor temperatures ($T_{\text{winding}} < 80^\circ\text{C}$), and monitor the hardwired Emergency Stop (E-Stop) loop. Upon bus packet dropout exceeding 10 ms or E-Stop assertion, firmware triggers dynamic regenerative braking and contactor isolation.

5.3 Mathematical Formulation of the Command Contract and Safety Projection

To guarantee operational safety without compromising cognitive adaptability, all commands flowing from Tier 1 and Tier 2 into Tier 3 must satisfy a formal *Command Contract*. Let the operational state of the robotic platform at time t be represented by the generalized state vector $\mathbf{x}(t) \in \mathcal{X}$, where:

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{p}_{\text{base}} \\ \theta_{\text{base}} \\ \mathbf{q}_{\text{arms}} \\ \mathbf{q}_{\text{grip}} \\ \mathbf{q}_{\text{head}} \end{bmatrix} \in SE(2) \times \mathbb{R}^{12} \times \mathbb{R}^2 \times \mathbb{R}^2, \quad (18)$$

and let $\mathbf{q} \in \mathcal{Q} \subset \mathbb{R}^{18}$ denote the complete active degree-of-freedom configuration vector (2 base drive wheels + 12 arm joints + 2 parallel grippers + 2 pan-tilt head joints).

When the cognitive VLA policy or external teleoperator dispatches an unconstrained configuration or Cartesian displacement delta $\Delta \mathbf{x}_{\text{AI}}$, this command cannot be directly committed to the actuator controllers. Instead, the admissible reference setpoint $\mathbf{x}_{\text{ref}}$ is computed via a constrained safety projection operator:

$$\mathbf{x}_{\text{ref}} = \Pi_{\mathcal{F}(\mathbf{q}, \mathcal{E}, \mathcal{S})}(\mathbf{x} + \Delta \mathbf{x}_{\text{AI}}), \quad (19)$$

where $\mathcal{F}(\mathbf{q}, \mathcal{E}, \mathcal{S}) \subset \mathcal{X}$ defines the time-varying, closed and convex set of verified physically feasible and collision-free operational states. The projection operator $\Pi_{\mathcal{F}}$ is mathematically defined as the constrained quadratic optimization:

$$\Pi_{\mathcal{F}}(\mathbf{y}) = \arg \min_{\mathbf{z} \in \mathcal{F}(\mathbf{q}, \mathcal{E}, \mathcal{S})} \frac{1}{2} \|\mathbf{z} - \mathbf{y}\|_{\mathbf{W}}^2 = \arg \min_{\mathbf{z} \in \mathcal{F}(\mathbf{q}, \mathcal{E}, \mathcal{S})} \frac{1}{2} (\mathbf{z} - \mathbf{y})^T \mathbf{W} (\mathbf{z} - \mathbf{y}), \quad (20)$$

where $\mathbf{W} \in \mathbb{R}^{m \times m}$ is a symmetric positive-definite weighting matrix prioritizing structural components (e.g., heavily penalizing base tipping while permitting compliant arm deflections).

5.3.1 Constrained Feasibility Set Formulation

The admissible manifold $\mathcal{F}(\mathbf{q}, \mathcal{E}, \mathcal{S})$ is formed by the intersection of four rigorous physical constraint sets:

$$\mathcal{F}(\mathbf{q}, \mathcal{E}, \mathcal{S}) = \mathcal{C}_{\text{kin}} \cap \mathcal{C}_{\text{stab}} \cap \mathcal{C}_{\text{env}}(\mathcal{E}) \cap \mathcal{C}_{\text{safe}}(\mathcal{S}). \quad (21)$$

1. **Kinematic and Velocity Bounds ($\mathcal{C}_{\text{kin}}$):** Enforces physical joint rotation limits, velocity saturation, and acceleration limits:

$$\mathcal{C}_{\text{kin}} = \left\{ \mathbf{x} = \mathbf{f}_{\text{kin}}(\mathbf{q}) \left| \begin{array}{l} \mathbf{q}_{\min} \leq \mathbf{q} \leq \mathbf{q}_{\max}, \\ -\dot{\mathbf{q}}_{\max} \leq \mathbf{J}^\dagger(\mathbf{q}) \dot{\mathbf{x}} \leq \dot{\mathbf{q}}_{\max}, \\ -\ddot{\mathbf{q}}_{\max} \leq \ddot{\mathbf{q}} \leq \ddot{\mathbf{q}}_{\max} \end{array} \right. \right\}, \quad (22)$$

where $\mathbf{f}_{\text{kin}}(\cdot)$ denotes the forward kinematics mapping, $\mathbf{J}(\mathbf{q})$ is the geometric manipulator Jacobian, and $\mathbf{J}^\dagger$ is its Moore-Penrose pseudoinverse.

2. **Dynamic Base Stability and Center of Mass Envelope ($\mathcal{C}_{\text{stab}}$):** To prevent chassis rollover during high-acceleration torso pitching or fast manipulator reaching with heavy payloads, the dynamic Zero-Moment Point (ZMP) $\mathbf{p}_{\text{ZMP}}$ must reside strictly within a contracted interior of the chassis support polygon $\mathcal{P}_{\text{support}}$ formed by the two 200 mm drive wheels and dual 100 mm casters:

$$\mathcal{C}_{\text{stab}} = \{ \mathbf{x} \mid \mathbf{p}_{\text{ZMP}}(\mathbf{q}, \ddot{\mathbf{q}}, \mathbf{a}_{\text{base}}) \in \text{int}_\delta(\mathcal{P}_{\text{support}}) \}, \quad (23)$$

where the planar ZMP is evaluated dynamically from the lumped mass distribution:

$$\mathbf{p}_{\text{ZMP}} = \frac{\sum_{i=1}^N m_i (\ddot{z}_i + g) \mathbf{r}_{i,xy} - \sum_{i=1}^N m_i \ddot{\mathbf{r}}_{i,xy} z_i}{\sum_{i=1}^N m_i (\ddot{z}_i + g)}, \quad (24)$$

with m_i and $\mathbf{r}_i = [\mathbf{r}_{i,xy}^T, z_i]^T$ representing the mass and Cartesian position of link i , $g = 9.81 \text{ m/s}^2$ the gravitational acceleration, and $\text{int}_\delta(\cdot)$ denoting a conservative safety margin contraction of $\delta = 35 \text{ mm}$ from the physical polygon boundary.

3. **Environmental Obstacle Clearance ($\mathcal{C}_{\text{env}}(\mathcal{E})$):** Based on real-time 3D Signed Distance Fields (SDF) and 2D Nav2 local costmaps generated from the environment perception state $\mathcal{E}$, every rigid body link $\mathcal{B}_k(\mathbf{q})$ must maintain a positive distance margin from external obstacles $\mathcal{O}_{\text{env}}$:

$$\mathcal{C}_{\text{env}}(\mathcal{E}) = \left\{ \mathbf{x} \left| \min_k \text{dist}(\mathcal{B}_k(\mathbf{q}), \mathcal{O}_{\text{env}}) \geq d_{\text{safe}} \right. \right\}, \quad (25)$$

where $d_{\text{safe}} = 0.05 \text{ m}$ for static workspace fixtures and $d_{\text{safe}} = 0.25 \text{ m}$ for dynamic human obstacles.

4. **Supervisory Safety State ($\mathcal{C}_{\text{safe}}(\mathcal{S})$):** Governed by discrete safety system variables $\mathcal{S}$:

$$\mathcal{C}_{\text{safe}}(\mathcal{S}) = \left\{ \mathbf{x} \left| \begin{array}{l} E_{\text{stop}} = \text{NORMAL}, \quad \text{SoC} \geq 10\%, \\ T_{\text{actuator}} \leq 80^\circ \text{C}, \quad \|\mathbf{F}_{\text{contact}}\| \leq F_{\text{threshold}} \end{array} \right. \right\}. \quad (26)$$

5.3.2 Anomaly Recovery and Observable Veto Semantics

Crucially, when the projection operator $\Pi_{\mathcal{F}}$ clamps an AI proposal, or when numerical infeasibility occurs, the command is *not* discarded silently. Let the projection residual be:

$$\varepsilon_{\text{proj}} = \|\Pi_{\mathcal{F}(\mathbf{q}, \mathcal{E}, S)}(\mathbf{x} + \Delta\mathbf{x}_{\text{AI}}) - (\mathbf{x} + \Delta\mathbf{x}_{\text{AI}})\|. \quad (27)$$

If $\varepsilon_{\text{proj}} > \varepsilon_{\text{tol}}$ (where $\varepsilon_{\text{tol}} = 0.04$ m in task space), Tier 3 immediately raises an observable **SafetyProjectionVeto** event over the ROS 2 middleware. This event halts downstream setpoint transmission, activates a smooth joint-space deceleration profile, and passes an execution failure signal to the Tier 2 **BehaviorTree.CPP** coordinator. The behavior tree responds by executing structured recovery nodes (e.g., retrying with an alternative grasp pose, repositioning the mobile base, or prompting the user via the Web Dashboard).

5.4 System-Level Mechatronic Interconnection Architecture

To guarantee electrical isolation, deterministic communication, and absolute human safety, the semi-humanoid platform is partitioned into four physically distinct interconnection networks:

1. **Power Distribution Network:** Galvanically isolates high-noise inductive actuator loads from sensitive digital compute. A central 24V 20Ah LiFePO₄ battery pack feeds two branches:
 - *Protected Logic Rail (Uncuttable 24V):* Routed directly from the BMS through slow-blow fuses to high-efficiency synchronous DC-DC buck converters supplying regulated 19V (Jetson Orin NX), 12V (head servos and cooling fans), and 5V (ESP32 controllers, sensors, and logic).
 - *Switched Motor Power Rail (Cutttable 24V):* Routed through a heavy-duty 24V 40A continuous safety contactor relay. This rail supplies the 9× CubeMars AK70-10 QDD actuators and the dual base BLDC drive motors.
2. **Deterministic Fieldbus Network (CAN 2.0B / TWAI at 1 Mbps):** A shared, high-speed differential bus interconnecting the Jetson compute host (via an isolated USB-to-CAN gateway) with the distributed ESP32 microcontrollers and the 9× AK70-10 QDD internal drivers. Employs 120Ω terminating resistors and shielded twisted-pair (STP) cabling.
3. **High-Bandwidth Perception and Telemetry Network:** High-speed point-to-point buses comprising Gigabit Ethernet (1000BASE-T) and USB 3.0/2.0 lines. Interconnects the Intel RealSense RGB-D camera, Slamtec RPLiDAR A1M8, ReSpeaker microphone array, Torso physical panel ports, and dual-band Wi-Fi 6 router.
4. **Hardwired Fail-Safe E-Stop Network:** An active-low, dual-channel physical loop connecting mechanical latching mushroom push-buttons (chassis-mounted and optional wireless tether) directly to the coil driver of the 40A safety contactor. Depressing an E-Stop button breaks coil power within ≤ 10 ms, physically cutting power to all high-torque actuators while maintaining complete power to compute, sensors, telemetry, and error logging.

Figure 5 depicts the system-level mechatronic block diagram, detailing the routing of all four independent physical networks across the platform subsystems.

6 Detailed Subsystems Mechatronic Design

6.1 Modular Device-Level Mechatronic Philosophy

In accordance with the VDI 2206 guidelines for cyber-physical mechatronic systems [6], the semi-humanoid robotic platform is partitioned into discrete, independently verifiable, and functionally encapsulated *devices*. Rather than treating the robot as a monolithic electro-mechanical assembly, each device integrates its own structural framework, specialized actuators, primary sensing transducers, localized embedded microcontroller (Espressif ESP32-WROOM-32) or central compute module, and standardized communication interfaces.

This modular device abstraction delivers three decisive engineering advantages:

1. **Isolated Unit Testing and Hardware-in-the-Loop (HIL) Verification:** Every subsystem can be powered, calibrated, dynamically tuned, and benchmarked on a standalone test bench before final physical integration onto the chassis.

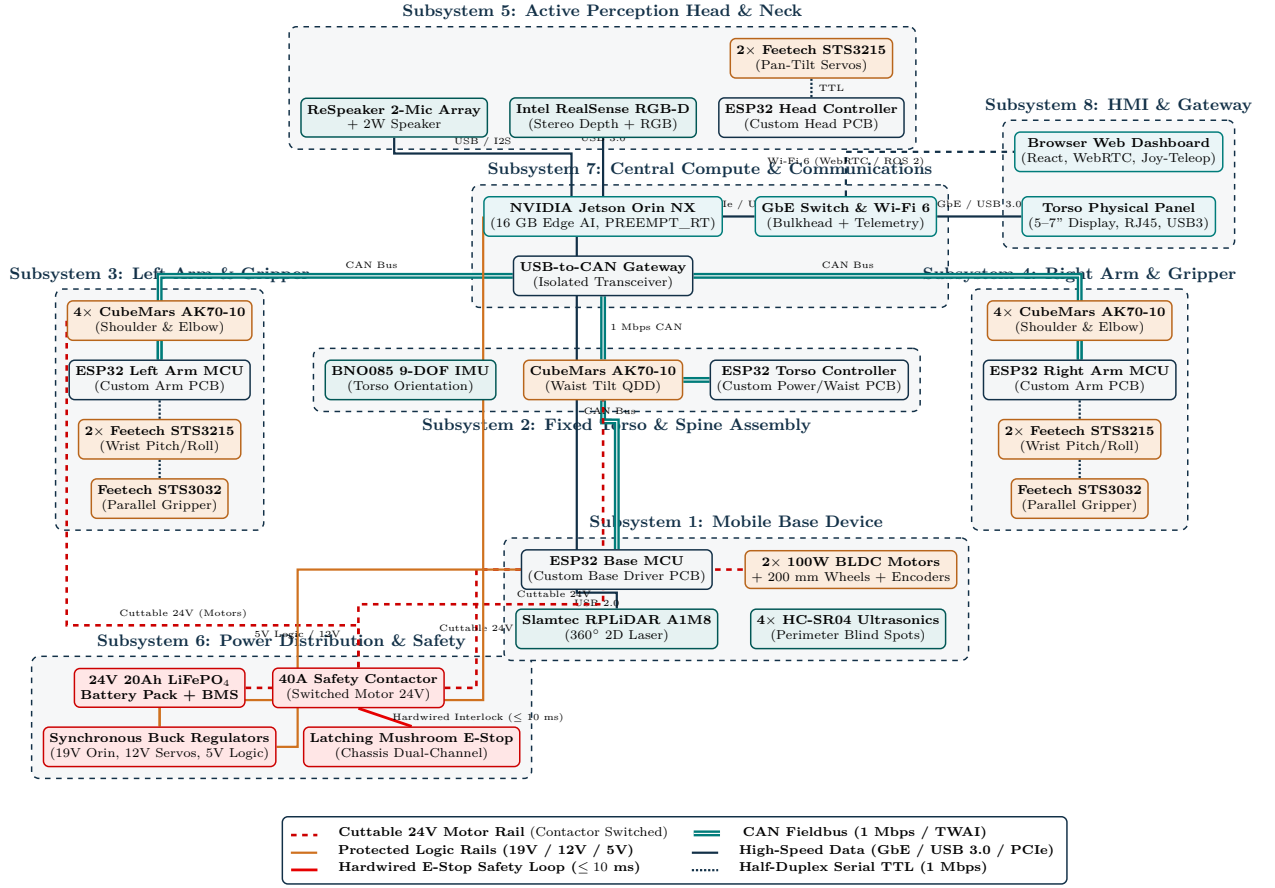


Figure 5: System-level mechatronic block diagram illustrating the four physically distinct interconnection networks: Switched Motor Power Rail, Protected Logic Power Rails, Deterministic CAN Fieldbus (1 Mbps), High-Speed Perception/Telemetry (GbE/USB 3.0), and the Hardwired Fail-Safe E-Stop Loop.

2. **Standardized Bus Architecture and Hot-Swappability:** Devices communicate across standardized physical links (1 Mbps CAN 2.0B / TWAI, half-duplex serial TTL, Gigabit Ethernet, and USB 3.0). Failed assemblies can be replaced rapidly without rewiring the central harness.
3. **Decoupled Software and Firmware Maintenance:** Microcontroller firmware manages low-level PID/FOC regulation natively, exposing standardized ROS 2 topics and action endpoints to Tier 2 and Tier 3 software.

Figure 6 illustrates the structural partitioning and logical boundaries across all eight core subsystems of the platform.

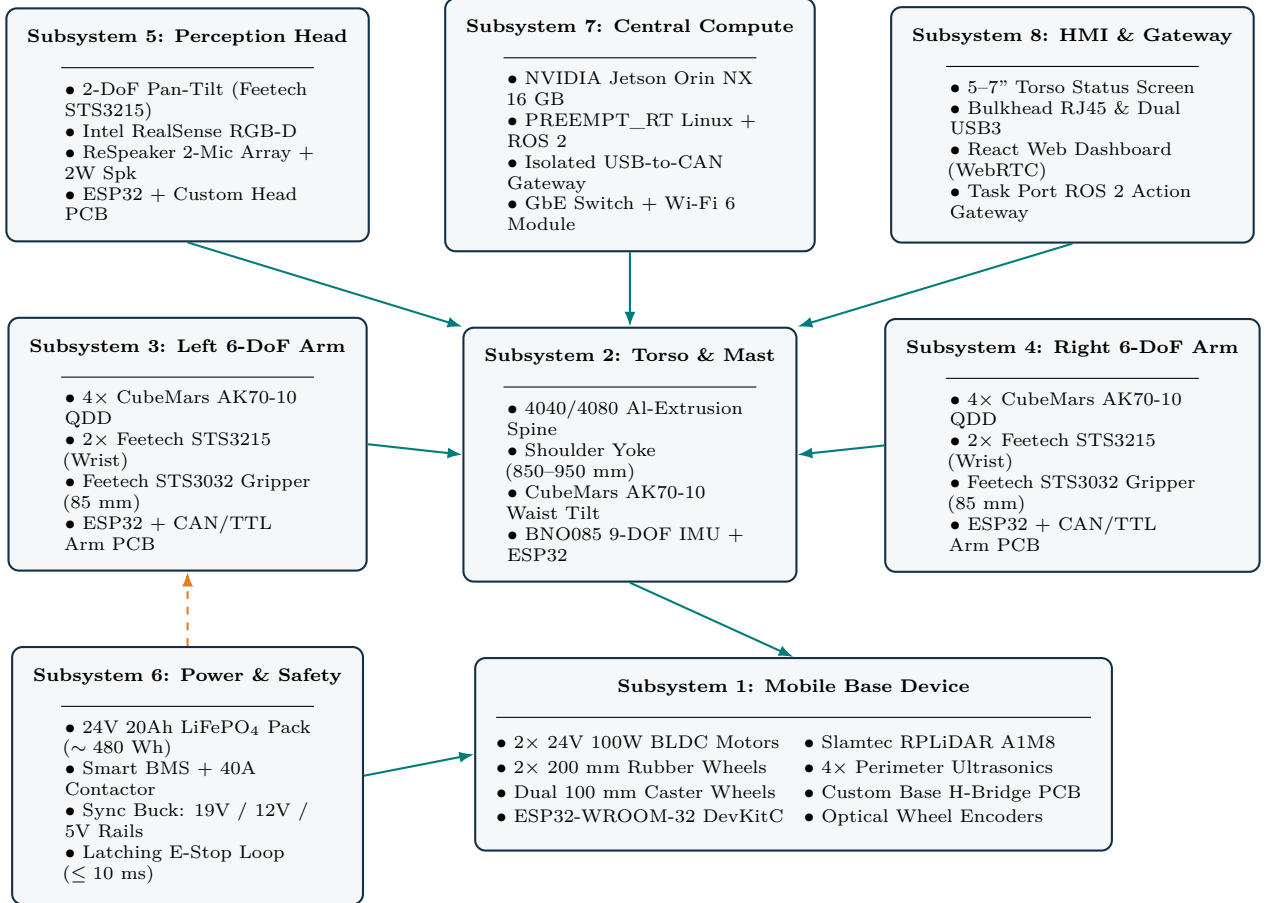


Figure 6: Device-level modular decomposition illustrating the encapsulation of mechanical structures, actuators, embedded electronics, and communications across all eight platform subsystems.

6.2 Subsystem 1: Mobile Base Device

The Mobile Base Device provides agile, high-traction, and stable indoor transit across unstructured domestic and industrial floor environments. Rather than utilizing complex, maintenance-intensive Mecanum wheels that suffer from severe slippage, vibration, and debris jamming on standard indoor carpeting, the platform employs a high-traction differential drive kinematic topology paired with dual passive support points.

6.2.1 Kinematics and Wheel Sizing Rationale

The drivetrain utilizes two active drive wheels ($D_{\text{wheel}} = 200$ mm, $b_{\text{wheel}} = 50$ mm) featuring high-grip vulcanized rubber treads, mounted coaxially with a track gauge of $L = 460$ mm. The large 200 mm wheel diameter was specifically selected to ensure seamless traversal over standard architectural thresholds (15–20 mm height), thick carpets, and floor-routed power cabling without chassis stalling.

Chassis balance and four-point stability are provided by two 100 mm heavy-duty load-rated swivel caster wheels placed symmetrically at the front and rear of the chassis perimeter. The total footprint is bounded

within 520 mm × 520 mm, comfortably satisfying Requirement G01 for navigating standard 750 mm interior doorways.

The mobile base velocity vector $\mathbf{v}_{\text{base}} = [v_x, \omega_z]^T$ maps directly to the left and right wheel angular velocities $[\dot{\theta}_L, \dot{\theta}_R]^T$ via standard differential drive kinematics:

$$\begin{bmatrix} \dot{\theta}_R \\ \dot{\theta}_L \end{bmatrix} = \frac{1}{R_{\text{wheel}}} \begin{bmatrix} 1 & L/2 \\ 1 & -L/2 \end{bmatrix} \begin{bmatrix} v_x \\ \omega_z \end{bmatrix}. \quad (28)$$

6.2.2 Actuation and Power Transmission

The drive wheels are powered by two 24V, 100W high-torque DC/BLDC gearmotors featuring integrated $\sim 30 : 1$ steel planetary gearboxes. At nominal speed (1.0 m/s), the wheel rotational velocity is $\omega_w = v/R_{\text{wheel}} = 1.0/0.1 = 10 \text{ rad/s} \approx 95.5 \text{ RPM}$. With the $30 : 1$ reduction, the motor core spins at approximately 2865 RPM, operating within its peak efficiency band ($> 82\%$).

The continuous wheel torque required to accelerate the 45 kg loaded platform at $a_{\text{max}} = 0.5 \text{ m/s}^2$ up a 5° ramp ($\alpha = 0.087 \text{ rad}$) with rolling friction coefficient $C_{rr} = 0.02$ is calculated as:

$$\begin{aligned} F_{\text{total}} &= mg \sin(\alpha) + mg C_{rr} \cos(\alpha) + ma_{\text{max}} \\ &= (45)(9.81) \sin(5^\circ) + (45)(9.81)(0.02) \cos(5^\circ) + (45)(0.5) \\ &= 38.48 \text{ N} + 8.79 \text{ N} + 22.50 \text{ N} = 69.77 \text{ N}. \end{aligned} \quad (29)$$

Divided across the two drive wheels and accounting for the 0.1 m radius and a transmission efficiency $\eta = 0.85$, the required gearbox output torque per wheel is:

$$\tau_{\text{wheel}} = \frac{F_{\text{total}} R_{\text{wheel}}}{2\eta} = \frac{(69.77)(0.1)}{2(0.85)} = 4.10 \text{ N} \cdot \text{m}. \quad (30)$$

The selected 100W planetary gearmotors provide up to $7.5 \text{ N} \cdot \text{m}$ continuous torque ($15.0 \text{ N} \cdot \text{m}$ peak stall), yielding an engineering safety factor $S_F = 1.83$, guaranteeing robust acceleration and ramp climbing.

6.2.3 Sensing Suite and Blind-Zone Elimination

Navigation and collision avoidance rely on a complementary dual-tier sensory arrangement:

- **Slamtec RPLiDAR A1M8:** Mounted at the leading edge of the chassis, this 2D laser scanner executes 360° planar distance ranging up to 12 m at a scanning frequency of 5.5–10 Hz (8000 measurement points/second). It supplies high-resolution obstacle slices to the Nav2 2D costmap.
- **4× HC-SR04 Ultrasonic Rangefinders:** Arrayed at 90° intervals around the chassis perimeter (0.02–4.0 m range). Because 2D LiDAR beams pass through transparent glass doors and cannot detect obstacles below the planar scanning beam (e.g., low table bases or pet toys), the ultrasonic sensors provide essential sonar reflection redundancy, halting base motion upon proximity violations ($d < 0.15 \text{ m}$).

6.2.4 Embedded Controller and Custom Base Driver PCB

Local wheel velocity control and sensor acquisition are governed by an Espressif ESP32-WROOM-32 DevKitC running FreeRTOS. An optocoupled quadrature decoder peripheral decodes motor encoder ticks (1024 pulses/rev on the motor shaft, yielding 30,720 counts/rev at the wheel, or 0.02 mm linear resolution).

The ESP32 is hosted on a Custom Base Controller PCB integrating:

- A dual-channel full H-bridge MOSFET power stage rated for 24V, 20A continuous per channel, equipped with low-side current shunt amplifiers.
- Optical isolators separating high-voltage motor switching noise from the ESP32 logic.
- Onboard synchronous buck regulation ($24\text{V} \rightarrow 5\text{V}$, 3A) powering the ultrasonic sensors and LiDAR motor.
- A 3.3V CAN transceiver wired to the ESP32 Two-Wire Automotive Interface (TWAI) peripheral, linking the base to the Tier 3 middleware at 200 Hz.

6.3 Subsystem 2: Fixed Torso and Structural Mast

The Torso Subsystem serves as the structural spine of the semi-humanoid robot, reacting dynamic arm forces, housing central electronics, and establishing anthropomorphic proportions.

6.3.1 Structural Aluminum Extrusion Spine and Shoulder Yoke

The structural backbone is fabricated from rigid 6061-T6 structural aluminum extrusions (combination of 40 mm × 40 mm and 40 mm × 80 mm heavy profiles) braced with CNC-machined 6 mm gusset plates. The mast extends vertically from the mobile base chassis to support an ergonomic shoulder yoke located at a nominal height of 850–950 mm above floor level. This shoulder height positions the dual arm end-effectors within natural reach of standard domestic tables (750 mm), kitchen counters (900 mm), and upper shelving units (1350 mm).

6.3.2 Waist Joint Actuation Allocation: CubeMars AK70-10 QDD

In the core baseline kinematic configuration, the torso functions as a rigid, fixed structural mast, providing maximum structural rigidity (K), completely eliminating dynamic mast deflection during high-speed reaching, lowering the system center of mass ($z_{\text{CoM}} \leq 350$ mm), and locking the active joint chain at 18 active DoF.

To provide modularity and long-term research flexibility, the structural frame and project Bill of Materials (Table 6) allocate and procure a CubeMars AK70-10 Quasi-Direct Drive (QDD) actuator within the Torso Subsystem. This actuator can be configured as an active 1-DoF waist tilt joint near the torso base featuring:

- A high-torque brushless permanent-magnet motor coupled to an integrated 10 : 1 single-stage planetary gearbox.
- A peak torque of 24.8 N · m (10.2 N · m continuous) with low reflected inertia ($N^2 = 100$).
- An integrated high-frequency FOC driver and dual 14-bit magnetic absolute encoders (measuring both motor rotor and gearbox output angles).

When enabled, the waist tilt joint provides a pitch articulation range of $\theta_{\text{waist}} \in [-15^\circ, +30^\circ]$, extending reach for floor-level pickup or pitching backward to counterbalance heavy forward arm extensions during payload transport.

6.3.3 Inertial Sensing and Thermal Management

Torso tilt feedback is monitored at 200 Hz by a BNO085 9-DOF IMU breakout incorporating an ARM Cortex-M0+ running Hillcrest Labs SensorHub fusion algorithms. The BNO085 outputs low-drift quaternion orientations ($\leq 1.5^\circ$ dynamic error), enabling active balance stabilization and tipping detection.

The torso houses the central compute module and power converters within an internal bay. An active forced-convection cooling kit—comprising an aluminum heatsink and a dual-ball-bearing PWM blower fan (28 CFM)—maintains internal temperatures below 55°C under peak computational and electrical workloads.

6.3.4 Optional Modular Telescopic Elevation Column Add-on

For operational environments requiring extended floor-to-ceiling reach (0–1600 mm), the platform design supports an optional, modular Telescopic Elevation Column add-on, budgeted independently in the project Bill of Materials (Table 1 of the BOM). This module mounts directly between the mobile base chassis and the torso mast, comprising:

- A heavy-duty 24V electric linear actuator providing 1500 N continuous thrust across a 300 mm stroke, equipped with an internal acme lead screw and dynamic power-off holding brake.
- Precision linear guide rails and dual ball-bearing carriages that isolate the actuator drive rod from cantilevered bending moments generated by arm manipulation.
- An ESP32-based Custom Column Driver PCB with an H-bridge driver and limit-switch interlocks.

Because the add-on is completely modular, it can be integrated into the mechanical assembly without modifying the core torso or drivetrain design.

6.4 Subsystems 3 & 4: Dual 6-DoF Manipulator Arm Devices

The platform incorporates dual anthropomorphic 6-DoF serial manipulators (Left Arm, Subsystem 3; Right Arm, Subsystem 4), engineered for safe physical interaction, high backdrivability, and low distal inertia.

6.4.1 Anthropomorphic 6-DoF Kinematic Topology

Each manipulator comprises six active revolute joints configured to match anthropomorphic human proportions:

1. **Joint 1 (Shoulder Pitch, $\pm 120^\circ$):** Elevates and depresses the arm in the sagittal plane.
2. **Joint 2 (Shoulder Roll, -30° to $+110^\circ$):** Abducts and adducts the arm in the frontal plane.
3. **Joint 3 (Shoulder Yaw, $\pm 90^\circ$):** Internal and external humeral rotation along the upper-arm axis.
4. **Joint 4 (Elbow Pitch, 0° to $+135^\circ$):** Flexes and extends the forearm.
5. **Joint 5 (Wrist Pitch, $\pm 90^\circ$):** Flexion and extension of the end-effector.
6. **Joint 6 (Wrist Roll, $\pm 180^\circ$):** Pronation and supination of the gripper assembly.

This configuration provides complete 6-DoF Cartesian pose accessibility ($\mathbf{x}_{EE} \in SE(3)$), with upper-arm and forearm link lengths of $L_1 = 280$ mm and $L_2 = 260$ mm, yielding a total spherical reach of $R_{\text{reach}} \approx 625$ mm (including the gripper).

6.4.2 Hybrid Actuation Architecture and Sizing Rationale

A critical engineering innovation of this platform is its *hybrid actuation architecture*, which strategically pairs Quasi-Direct Drive (QDD) actuators at the high-load proximal joints with compact serial bus servos at the distal wrist joints:

Proximal Joints (Shoulder 3-DoF + Elbow 1-DoF): Joints 1 through 4 are actuated by CubeMars AK70-10 QDD modules (24.8 N · m peak torque, 10 : 1 planetary ratio, 4 units per arm, 8 units total across both arms). The low 10 : 1 reduction ratio limits reflected rotor inertia:

$$J_{\text{reflected}} = N^2 J_{\text{rotor}} = 100 \cdot J_{\text{rotor}}, \quad (31)$$

rendering the joints mechanically backdrivable. This backdrivability permits transparent proprioceptive torque estimation directly from motor phase currents:

$$\tau_{\text{est}} = N K_\tau I_q - \tau_{\text{friction}}(\dot{q}), \quad (32)$$

where K_τ is the motor torque constant (0.095 N · m/A) and I_q is the active quadrature phase current. This enables high-fidelity compliance and collision detection without requiring delicate, expensive 6-axis wrist force-torque transducers.

Distal Joints (Wrist Pitch + Wrist Roll): Joints 5 and 6 are driven by Feetech STS3215 smart serial bus servos (12V, 2.9 N · m stall torque, metal planetary gear train, contactless 12-bit magnetic absolute encoder, half-duplex TTL interface).

The engineering rationale for this hybrid division is governed by cantilever dynamics:

$$\tau_{\text{shoulder}} = \sum_{k=1}^6 m_k g r_{k,x} + m_{\text{payload}} g R_{\text{reach}}. \quad (33)$$

If heavy QDD actuators ($m_{\text{AK70}} \approx 520$ g) were utilized at the wrist, their mass would generate an unproductive gravitational moment at the shoulder of:

$$\Delta \tau_{\text{shoulder}} = 2 \cdot (0.52 \text{ kg}) \cdot (9.81 \text{ m/s}^2) \cdot (0.54 \text{ m}) \approx 5.51 \text{ N} \cdot \text{m}, \quad (34)$$

consuming more than 22% of the shoulder actuator’s peak capacity solely to support its own wrist motors. By deploying lightweight Feetech STS3215 servos ($m_{\text{STS}} \approx 55$ g), distal mass is reduced by nearly 90%, freeing substantial payload capacity (2.0–3.0 kg) and lowering arm actuation costs by more than 75%.

6.4.3 Embedded Architecture and Dual-Bus Arm Joint Interface PCB

Each arm is coordinated by a dedicated ESP32-WROOM-32 DevKitC mounted on a Custom Arm Joint Interface PCB. As shown in Figure 7, the interface PCB bridges two physically disparate communication buses:

1. **1 Mbps Differential CAN Bus (TWAI):** Connects the ESP32 to the daisy-chained internal FOC drivers of the four AK70-10 QDD actuators. Employs a 3.3V CAN transceiver (SN65HVD230) and split 120Ω termination resistors. The ESP32 dispatches synchronous cyclic torque/position frames at 1 kHz.
2. **1 Mbps Half-Duplex Serial TTL Bus:** Driven by the ESP32 UART2 peripheral via a 74HC126 tri-state buffer circuit. This single-wire differential line controls the two STS3215 wrist servos and the STS3032 gripper servo in a daisy-chain configuration, polling 12-bit positions and current feedback at 200 Hz.

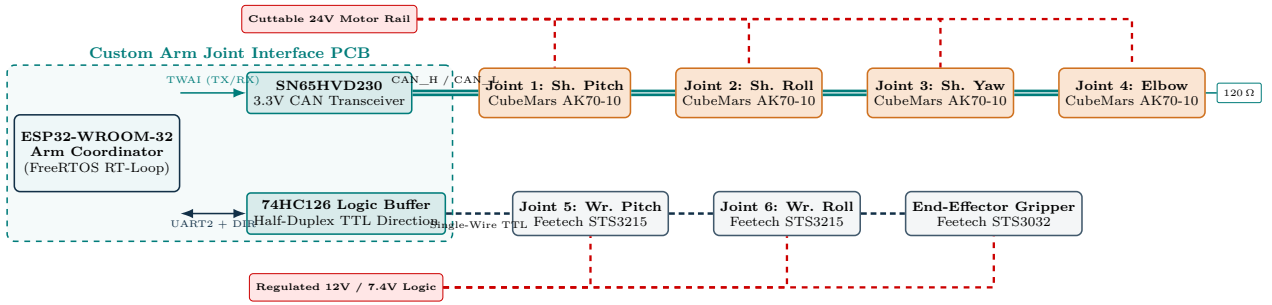


Figure 7: Arm hybrid actuation and dual-bus control topology. The ESP32-WROOM-32 coordinator simultaneously drives the 1 Mbps differential CAN bus (controlling four CubeMars AK70-10 QDD actuators) and a 1 Mbps half-duplex serial TTL line (controlling two Feetech STS3215 wrist servos and the STS3032 gripper servo).

6.5 End-Effector: 2-Finger Parallel Gripper

Both arms terminate in a lightweight, modular 2-finger parallel-jaw gripper engineered for everyday object manipulation.

6.5.1 Mechanics and Linear Stroke

The gripper mechanism utilizes a symmetrical dual-rack-and-pinion transmission driven directly by a Feetech STS3032 compact serial bus servo (7.4V, 4.5 N · m stall torque). The pinion engages two opposing hardened nylon racks guiding the fingers along miniature stainless steel linear rails, delivering a linear jaw stroke of:

$$S_{\text{stroke}} = 0\text{--}85 \text{ mm}, \quad (35)$$

with a full closing duration of $t_{\text{close}} \approx 0.35 \text{ s}$. The 85 mm stroke accommodates the vast majority of domestic items, including drinking glasses, bottles, food cans, fruits, and small boxes.

6.5.2 Compliant Silicone Gripping Pads

The rigid 3D-printed fingers (Carbon-Fiber Reinforced PETG) are fitted with modular, vulcanized silicone pads featuring high-friction herringbone tread patterns ($\mu_{\text{friction}} \geq 0.85$ on dry glass/metal). The compliant silicone layer (3 mm thickness, Shore 40A durometer) passively conforms to curved and irregular workpiece geometries, distributing contact pressures and preventing object slippage under dynamic arm accelerations.

6.5.3 Current-Based Grasp Force Regulation

The gripper achieves force-sensitive grasping without delicate external strain-gauge load cells by reading motor phase current feedback from the STS3032 servo:

$$F_{\text{grip}} \approx \frac{K_{\tau, \text{servo}} \cdot I_{\text{servo}}}{r_{\text{pinion}}}, \quad (36)$$

where $r_{\text{pinion}} = 12$ mm is the pitch radius of the drive gear. The ESP32 firmware monitors current draw at 200 Hz. Upon detecting a steep current gradient indicating object contact, the controller clamps jaw motion to a programmable force setpoint ($F_{\text{target}} \in [5 \text{ N}, 25 \text{ N}]$), preventing the crushing of fragile objects (e.g., disposable plastic cups or ripe fruit).

6.6 Subsystem 5: Active Perception Head and Neck Device

The Active Perception Head acts as the visual and auditory sensory hub of the platform, supporting autonomous navigation, spatial manipulation, and human-robot dialogue.

6.6.1 2-DoF Pan-Tilt Kinematics

The head assembly is mounted atop the torso mast via a 2-DoF active pan-tilt gimbal mechanism driven by two Feetech STS3215 12V serial bus servos (2.9 N · m stall torque). The kinematic range provides:

- **Pan Axis:** $\theta_{\text{pan}} \in [-90^\circ, +90^\circ]$ (horizontal azimuthal sweep).
- **Tilt Axis:** $\theta_{\text{tilt}} \in [-45^\circ, +30^\circ]$ (downward workspace inspection to upward face tracking).

Because the entire sensor head payload (camera, microphones, speaker, and housing) weighs less than 450 g, the STS3215 servos operate far below their thermal limits, achieving slew velocities up to $180^\circ/\text{s}$ for rapid saccadic gaze shifts.

6.6.2 Vision and Acoustic Transducers

- **Intel RealSense RGB-D Camera:** Streams active infrared stereo depth (1280×720 at 30 fps) and full-color RGB frames via a USB 3.0 link directly to the Jetson Orin NX host. Operating over a range of 0.2–3.0 m, it feeds both the 3D voxel TSDF reconstruction pipeline and the VLA policy visual encoder.
- **ReSpeaker 2-Mic Pi HAT Array:** Incorporates dual omnidirectional microphones with an onboard stereo codec. Features hardware-accelerated acoustic echo cancellation (AEC) and Time-Difference-of-Arrival (TDOA) sound localization, allowing the head to turn autonomously toward speaking humans.
- **2W Broad-Dispersion Speaker:** Driven by an onboard I2S Class-D audio amplifier, delivering clear synthesized speech dialogue, task feedback, and audible emergency alarms.

6.6.3 Custom Head Interface PCB

An ESP32-WROOM-32 DevKitC and a Custom Head Interface PCB manage local pan-tilt servo commands, decode I2S audio streams, and drive an external addressable RGB LED indicator ring (displaying robot emotional state and system operational modes).

6.7 Subsystem 6: Power Distribution and Safety Subsystem

The Power Distribution and Safety Subsystem delivers clean, protected electrical energy across all active platform devices while enforcing fail-safe hardware disconnection.

6.7.1 Battery Chemistry and Energy Capacity

The primary electrical power source is a high-energy-density 24V, 20Ah Lithium Iron Phosphate (LiFePO_4) battery pack (~ 480 Wh nominal energy storage). Compared to standard Lithium-Ion (NMC) cells, LiFePO_4 chemistry offers superior thermal runaway resistance (thermal threshold $> 270^\circ\text{C}$), an extended operational cycle life (> 2000 cycles at 80% DoD), and high continuous discharge capability ($2C = 40\text{A}$).

At a nominal continuous platform power draw of 115 W (35W compute + 30W idle actuators + 50W base transit), the 480 Wh pack guarantees an uninterrupted operational runtime of:

$$t_{\text{runtime}} = \frac{480 \text{ Wh} \cdot 0.85}{115 \text{ W}} \approx 3.55 \text{ hours}, \quad (37)$$

fully fulfilling Requirement E02. The pack is paired with a dedicated 24V, 5A LiFePO_4 profile smart charger providing full recharge in approximately 4 hours.

6.7.2 Galvanic Partitioning and Synchronous DC-DC Regulation

To eliminate ground loops and prevent actuator electromagnetic interference (EMI) from corrupting high-speed digital buses, power distribution is split across two architecturally separated domains:

1. **Protected Logic Rail (Uncuttable 24V):** Feeds three high-efficiency ($> 94\%$) synchronous buck regulators:
 - 24V $\rightarrow$ 19V (6A, 114W max): Supplies regulated power to the NVIDIA Jetson Orin NX host.
 - 24V $\rightarrow$ 12V (10A, 120W max): Powers the Feetech serial bus servos, audio amplifiers, and cooling blowers.
 - 24V $\rightarrow$ 5V (10A, 50W max): Supplies the distributed ESP32 microcontrollers, IMU, ultrasonic sensors, and LiDAR motor.
2. **Switched Motor Power Rail (Cuttable 24V):** Supplies raw battery voltage directly to the $9\times$ CubeMars AK70-10 QDD actuators and the dual base BLDC gearmotors through a heavy-duty 24V, 40A continuous safety contactor relay.

6.7.3 Hardware Emergency Stop (E-Stop) Disconnect Circuit

Operational safety compliance (ISO 13482 [5]) is guaranteed via a dedicated, hardwired dual-channel safety circuit. A red latching mushroom E-Stop push-button is mounted prominently on the upper torso spine, wired in series with an optional wireless safety receiver. Depressing the E-Stop button breaks the coil circuit of the 40A safety contactor. Spring-loaded mechanical contacts disconnect the motor power rail in:

$$t_{\text{disconnect}} \leq 8 \text{ ms} \ll 10 \text{ ms} \quad (\text{Requirement E03}). \quad (38)$$

Because the logic rail bypasses the contactor, compute, perception, wireless telemetry, and system diagnostic logs remain fully energized, allowing the robot to transmit immediate fault diagnostics without losing system state.

6.8 Subsystem 7: Central Compute and Communications

Subsystem 7 centralizes high-level artificial intelligence inference, real-time trajectory generation, and multi-interface communications.

6.8.1 Primary Edge Compute: NVIDIA Jetson Orin NX 16 GB

High-level computational processing is performed by an NVIDIA Jetson Orin NX module (16 GB 128-bit LPDDR5 RAM, 1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores, and an 8-core Arm Cortex-A78AE v8.2 64-bit CPU). The module delivers up to 100 TOPS of sparse INT8 AI performance, comfortably accommodating parallel execution of:

- Quantized multimodal VLA models (OpenVLA / π_0) running via TensorRT-LLM at 5–10 Hz.
- Real-time 3D Signed Distance Field (SDF) voxel mapping and depth-point registration.
- 2D SLAM and MPPI local trajectory generation at 50 Hz.

6.8.2 Operating System and Real-Time Scheduling Architecture

The host runs Ubuntu 22.04 LTS (JetPack 6) patched with the PREEMPT_RT real-time kernel. To eliminate thread contention and scheduling jitter, computational resources are partitioned using Linux Control Groups (cgroups) and CPU core affinity pinning:

- **Cores 0–1 (Real-Time Middleware):** Isolated from standard Linux scheduling; dedicated exclusively to `ros2_control`, fieldbus CAN I/O threads, and the 200 Hz state estimation UKF running under `SCHED_FIFO` priority.
- **Cores 2–5 (Mission Autonomy):** Executes Nav2 costmap processing, MoveIt 2 collision checking, and BehaviorTree.CPP evaluation.
- **Cores 6–7 (Background Services):** Hosts the web server, WebRTC streaming pipeline, rosbridge WebSocket endpoints, and system logging.

6.8.3 Communication Gateways and Network Infrastructure

- **Isolated USB-to-CAN Gateway:** Connects the Jetson host to the platform’s primary 1 Mbps CAN bus, featuring optical galvanic isolation (2.5 kV_{RMS}) and hardware message timestamping ($< 10 \mu\text{s}$).
- **Industrial Gigabit Network Switch:** Routes internal 1000BASE-T Ethernet between the Jetson host, external torso bulkhead RJ-45 service ports, and high-speed telemetry bridges.
- **Dual-Band Wi-Fi 6 & Bluetooth 5.2:** Provides high-throughput, low-latency wireless communication ($> 600 \text{ Mbps}$) for remote browser-based teleoperation, cloud fleet management, and remote software deployment.

6.9 Subsystem 8: HMI, Web Dashboard, and Task Port Gateway

The Human-Machine Interface (HMI) establishes intuitive physical, browser-based, and algorithmic interfaces for operating and programming the semi-humanoid robot.

6.9.1 Physical Torso Panel and Bulkhead Ports

Mounted on the front and rear of the upper torso housing, the physical panel provides immediate local diagnostics and field service access:

- **5–7” Touchscreen Status Display:** Presents high-contrast system telemetry, including current Wi-Fi SSID and assigned IP address, battery State of Charge (SoC%), active ROS 2 node states, actuator temperatures, and active error codes.
- **Bulkhead Programming and Service Bay:** Features an industrial RJ-45 Gigabit Ethernet port, dual high-speed USB 3.0 service ports, a keyed master power switch, and a soft reboot button. This allows developers to flash microcontrollers, extract flight-recorder logs, and deploy software updates without dismantling chassis panels.

6.9.2 Browser Web Dashboard and Teleoperation Console

Remote teleoperation and fleet monitoring are conducted via a responsive, single-page web dashboard developed using React, TypeScript, and TailwindCSS, communicating with the robot over WebRTC and WebSocket links:

- **Low-Latency Teleoperation Console:** Streams low-latency ($< 80 \text{ ms}$) H.264 video from the RealSense camera via WebRTC. Operators can command mobile base translation and arm end-effector Cartesian jogging using virtual on-screen joysticks, standard keyboard hotkeys, or USB game controllers (via the HTML5 Gamepad API).
- **Diagnostic Alarm Console:** Displays real-time visual alerts and audible warnings upon anomalous hardware states, such as actuator overtemperature ($T > 75^\circ\text{C}$), fieldbus packet loss, battery under-voltage (SoC $< 15\%$), or LiDAR point cloud occlusion.
- **Interactive Task Programming (Macro Waypoint Recorder):** Enables non-expert operators to program complex mobile manipulation tasks graphically. The interface features a record-and-playback macro recorder: operators lead the compliant arm through a sequence of waypoints, save Cartesian and gripper states as named skills, and link them into executable behavior tree routines.

6.9.3 Task Port Gateway: Decoupled ROS 2 Action Servers

The Task Port Gateway implements the software isolation boundary described in Section 5. It encapsulates low-level hardware complexity behind standardized ROS 2 Action servers:

- **Navigation Action:** `/navigate_to_pose` dispatches $SE(2)$ goals to Nav2 (action type: `NavigateToPose`).
- **Manipulation Actions:** `/pick_object` and `/place_object` command grasp and placement routines (action type: `PickObject`).
- **Autonomous Docking Action:** `/dock_robot` coordinates contact engagement with the dock (action type: `Dock`).

By forcing all cognitive models and external software pipelines to interact strictly through these typed Action endpoints, the Task Port Gateway guarantees that unexpected high-level crashes or network disconnects never jeopardize the physical integrity of the platform.

7 Cross-Domain Engineering Allocation

7.1 Multidisciplinary Allocation Framework (VDI 2206)

The realization of an autonomous mobile manipulator demands a rigorous departure from fragmented, discipline-isolated development methodologies. In conventional engineering workflows, the mechanical skeleton, electrical power grid, motor actuation, and software intelligence are frequently addressed in consecutive isolation. For high-degree-of-freedom cyber-physical systems (CPS), this serialized paradigm routinely induces structural inefficiencies, thermal runaway at motor joints, communication bus saturation, and costly redesign iterations.

To systematically overcome these failure modes, the system-level conceptual requirements (R_1 – R_{24}) and tripartite flow structures (**M**, **E**, **I**) formulated in Section 4 are translated into domain-specific engineering work packages in strict accordance with the VDI 2206 design methodology [6]. Operating across the descending and ascending branches of the VDI 2206 V-model, the platform architecture establishes a synchronized triad across three primary engineering domains:

1. **Mechanical Engineering:** Structural chassis, kinematics linkages, mass/inertia distribution, thermal heat sinking, and environmental ingress protection.
2. **Electrical & Embedded Engineering:** High-current power distribution, Li-ion battery management, safety interlocks, custom PCB design, and hard real-time motor commutation firmware.
3. **Computer Science and Software Engineering:** Real-time middleware orchestration, SLAM and navigation, collision-free manipulation planning, Vision-Language-Action (VLA) foundation models, and human-machine teleoperation interfaces.

Table 5 formalizes the domain allocation matrix, detailing the division of responsibilities, primary work packages, concrete deliverables, and quantifiable verification metrics across the platform lifecycle.

Table 5: Formal VDI 2206 cross-domain engineering allocation matrix for the semi-humanoid platform.

Domain	Work Package (WP)	Scope & Key Responsibilities	Primary Deliverables	Verification Metric
Mechanical	WP-ME1: Base Drivetrain & Structural Chassis	Differential chassis frame, motor mounts, wheel hubs, ground clearance (> 30 mm), and caster suspension.	3D CAD assembly, structural FEA stress reports, CNC fabrication drawings.	Factor of safety $S_f \geq 2.2$; payload ≥ 80 kg; chassis deflection < 0.5 mm.
	WP-ME2: Fixed Mast Torso & Modular Waist Axis	Structural fixed mast frame, CubeMars AK70-10 modular mounting, bearing load isolation, and CoM balance.	Torso weldment/bracket CAD, bearing load calculation sheets, physical frame.	Structural deflection < 0.5 mm; modular waist moment load ≥ 45 N · m.
	WP-ME3: Dual 6-DoF Manipulator Kinematic Chain	Shoulder, elbow, and wrist linkages, link inertia reduction, carbon-fiber / aluminum structural tubes, cable passages.	Parametric arm CAD, mass/inertia matrices (I_{xx}, I_{yy}, I_{zz}), 3D printed / machined links.	Natural frequency $\omega_n \geq 18$ Hz; settling time $t_s \leq 0.4$ s; arm payload ≥ 2.0 kg.
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Table 5 continued from previous page

Domain	Work Package (WP)		Scope & Responsibilities	Key	Primary Deliverables	Verification Metric
	WP-ME4: Modular End-Effectors & 2-DoF Neck		Parallel-jaw grippers with silicone contact pads, 2-DoF pan-tilt sensor head housing RGB-D camera and microphones.		Gripper CAD with ISO 9409-1 flange, pan-tilt head gimbal assembly, 3D MJF PA12 parts.	Clamping force 15–35 N; grasp span 0–85 mm; head pan $\pm 90^\circ$, tilt $+45^\circ / -30^\circ$.
Electrical &	WP-EE1: Subsystem & Interlocks	Power Safety	24V, 20Ah LiFePO ₄ battery integration, BMS, main 40 A contactor, E-Stop disconnect loop, and multi-rail DC-DC.		Power schematic, wiring harness pinout diagrams, battery enclosure, safety contactor board.	E-Stop disconnect latency ≤ 10 ms; voltage ripple $< 1.5\%$; battery run-time ≥ 4.0 h.
Embedded	WP-EE2: Subsystem PCB Design	Custom	Design and population of 4 custom boards: Base Controller, Power/Waist, Head Interface, and Dual Arm Joint Interface.		KiCad schematics, multi-layer Gerber files, assembled and bench-tested PCBs.	100% net continuity; CAN bus termination $120\ \Omega \pm 1\%$; 40 A trace thermal rise $< 25^\circ\text{C}$.
	WP-EE3: Embedded Firmware	Real-Time	Firmware for 5x ESP32-WROOM-32 MCUs, TWAI CAN drivers, CiA 402 drive state machines, half-duplex TTL servo chains.		C/C++ FreeRTOS firmware repositories, micro-ROS client nodes, flash automation scripts.	Deterministic loop rate 1 kHz; CAN packet transmission jitter $< 150\ \mu\text{s}$; zero watchdog timeouts.
	WP-EE4: Conditioning & Harnessing	Signal	Quadrature encoder decoding, 9-DoF IMU SPI acquisition, ultrasonic ranging, continuous-rotation wire shielding.		Finished wiring harnesses with strain relief, ferrite choke clamp documentation.	EMI noise margin > 20 dB; zero dropped encoder pulses up to 3000 RPM; IMU rate ≥ 200 Hz.
Software &	WP-CS1: Infrastructure & Middleware	Compute RTOS	NVIDIA Jetson Orin NX OS setup, PREEMPT_RT real-time Linux kernel, ROS 2 Jazzy infrastructure, micro-ROS agent.		Custom Linux rootfs image, Docker deployment containers, ros2_control hardware interfaces.	Kernel interrupt latency $\leq 50\ \mu\text{s}$; inter-process zero-copy transport latency $< 100\ \mu\text{s}$.
Computer	WP-CS2: Autonomous Base Navigation & SLAM		2D LiDAR SLAM, costmap generation, dynamic obstacle avoidance, and Nav2 path tracking controllers.		Nav2 configuration packages, customized costmap plugins, AMCL tuning parameters.	Global localization error ≤ 30 mm, heading error $\leq 2.0^\circ$; transit speed 0.8–1.0 m/s.

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Table 5 continued from previous page

Domain	Work Package (WP)	Scope & Responsibilities	Key	Primary Deliverables	Verification Metric
Science	WP-CS3: Manipulation Kinematics & Dual-Arm Planning	MoveIt 2 integration, bio-IK / TRAC-IK inverse kinematics solvers, collision-free Cartesian trajectory generation.		MoveIt configuration packages, dual-arm collision models, self-collision avoidance matrices.	2 IK solve time ≤ 5 ms; Cartesian trajectory execution repeatability $\leq \pm 2.0$ mm.
	WP-CS4: Standardized Task Port & Safety Supervisor	Action servers exposing decoupled task primitives (/navigate_to_pose, /pick_object), runtime safety state observer.		ROS 2 Action Server packages, ISO 13482 safety monitoring node, Python client verification scripts.	Task dispatch latency ≤ 25 ms; safety override reaction time ≤ 20 ms; 100% command isolation.
	WP-CS5: Multimodal Perception & VLA Integration	RealSense RGB-D point cloud segmentation, Whisper speech recognition, OpenVLA foundation policy trajectory streaming.		Perception ROS 2 nodes, Whisper STT inference pipeline, OpenVLA action chunking adapter.	Object pose estimation rate ≥ 15 Hz; VLA action inference rate ≥ 10 Hz; speech response < 1.2 s.
	WP-CS6: Web Teleoperation & Fleet Telemetry	Browser-based dashboard providing real-time video streaming, 3D LiDAR point cloud display, and joint teleoperation.		Full-stack web application (React, WebRTC, rosbridge_suite WebSocket server).	Glass-to-glass video streaming latency ≤ 120 ms; telemetry update rate ≥ 20 Hz.

7.2 Domain Work Packages, Primary Deliverables, and Verification Metrics

7.2.1 Mechanical Engineering Domain

The mechanical engineering team holds primary ownership over the structural integrity, kinematics geometry, payload capacity, and mass properties of the robotic system:

- **Chassis and Drivetrain (WP-ME1):** Responsible for the structural baseplate machined from 6 mm 6061-T6 aluminum alloy. The chassis houses two 100 W brushed DC gearmotors driving $\varnothing 200$ mm high-traction rubber wheels, paired with dual 100 mm heavy-duty polyurethane swivel casters. Finite element analysis (FEA) validates that under a combined static and dynamic payload of 85 kg, von Mises stresses remain well below the yield strength ($\sigma_{\text{yield}} = 276$ MPa), yielding a safety factor $S_f \geq 2.2$.
- **Fixed Mast Torso Structure and Waist Axis (WP-ME2):** The baseline torso adopts a rigid structural fixed mast constructed from modular 30×30 mm aluminum extrusions, delivering the invariant 18-DoF baseline. To support the modular torso articulation path budgeted in the BOM, it incorporates precision mounting provisions and dual angular contact ball bearings for a CubeMars AK70-10 QDD waist tilt actuator, isolating the gearbox from parasitic radial and axial moment loads during forward reaches.
- **Dual 6-DoF Manipulators (WP-ME3):** Each 6-DoF arm is configured with four proximal CubeMars AK70-10 QDD actuators (shoulder pitch, shoulder roll, shoulder yaw, elbow pitch) and two distal Feetech STS3215 serial bus servos (wrist pitch, wrist roll). Structural links are engineered using thin-walled aluminum tubing and hollow carbon-fiber composite sections, minimizing link mass while shifting structural resonance frequencies above 18 Hz to avoid control-loop excitation.

- **Modular End-Effector and Head Assembly (WP-ME4):** The parallel-jaw gripper uses a compact Feetech STS3032 bus servo driving a dual-rack-and-pinion transmission. Gripper jaws feature exchangeable shore-A 60 silicone pads optimized for domestic tableware grasping (0–85 mm opening span). The perception head provides 2-DoF pan/tilt articulation driven by dual Feetech STS3215 servos, carrying the Intel RealSense depth camera and microphone array.
- **Optional Prismatic Column (WP-ME5):** When installed as a modular add-on, the telescopic column utilizes a 24 V electric linear actuator (300 mm stroke, 1500 N static thrust) guided by dual precision recirculating ball linear guide rails (MGN15 series), completely decoupling moment loads from the actuator lead screw.

7.2.2 Electrical and Embedded Engineering Domain

The electrical engineering team manages the complete energy transformation pipeline, power conditioning, circuit design, and distributed real-time microcontroller network:

- **Power Architecture and Energy Storage (WP-EE1):** The primary energy source is an 8S, 24V, 20Ah LiFePO₄ battery pack (~ 480 Wh) with an internal Battery Management System (BMS) supporting continuous 40A discharge and 100A transient pulse capability. A 40 A continuous-duty safety contactor provides galvanically isolated hardware power cutoff.
- **Custom PCB Hardware Development (WP-EE2):** To eliminate cabling rats-nests and mitigate ground loops, four custom populated PCBs are developed:
 1. *Custom Base Controller PCB:* Combines an ESP32 carrier, dual-channel H-bridge motor driver (40 V / 15 A per channel), optical encoder quadrature counters, and ultrasonic sensor headers.
 2. *Custom Power Distribution & Waist PCB:* Houses high-current fused battery busbars, three high-efficiency synchronous buck converters ($24\text{V} \rightarrow 19\text{V } 4\text{A}$, $24\text{V} \rightarrow 12\text{V } 6\text{A}$, and $24\text{V} \rightarrow 5\text{V } 5\text{A}$), a 3.3 V CAN transceiver (SN65HVD230) interfaced to the ESP32 Two-Wire Automotive Interface (TWAI), and safety relay coil drivers.
 3. *Custom Head Interface PCB:* Integrates half-duplex TTL bus servo level shifters, an I2S stereo class-D audio power amplifier (2×3 W), microphone breakout headers, and an ESP32 carrier.
 4. *Custom Arm Joint Interface PCBs (x2):* Mounted at each shoulder base, each board integrates a 3.3 V CAN transceiver on the ESP32 TWAI bus (interfacing the 4x AK70-10 QDD chain) and a 74HC126-based half-duplex TTL buffer circuit (interfacing the 2x wrist servos and 1x gripper servo).
- **Distributed Real-Time Firmware (WP-EE3):** Five ESP32-WROOM-32 microcontrollers execute FreeRTOS firmware. Low-level control tasks operate deterministically:
 - Base MCU: Executes closed-loop PID velocity tracking (100 Hz) using quadrature encoder feedback.
 - Torso MCU: Reads 9-DoF IMU data via SPI (200 Hz), executes waist angle trajectory interpolation, and monitors battery bus voltages.
 - Arm MCUs (x2): Translates high-level joint trajectory setpoints into 1 kHz CAN command frames dispatched to the CubeMars AK70-10 FOC inverters and 50 Hz serial packets dispatched to the Feetech bus servos.
 - Head MCU: Regulates pan/tilt tracking angles and manages audio input/output buffers.
- **Signal Integrity and Harnessing (WP-EE4):** Critical communication lines (CAN, UART, I2S) are routed through twisted shielded pairs with common-mode choke filtering to maintain robust signal margins in the vicinity of pulse-width modulated (PWM) motor power lines.

7.2.3 Computer Science and Software Engineering Domain

The software engineering team owns the high-level computational pipelines, real-time operating system middleware, spatial navigation, manipulation planning, and physical AI execution:

- **Operating System and Middleware Core (WP-CS1):** The NVIDIA Jetson Orin NX (16 GB) executes Ubuntu 24.04 LTS patched with the PREEMPT_RT real-time kernel. The core communication layer runs ROS 2 Jazzy Jalisco. Fast DDS serves as the default middleware implementation, configured with shared-memory (SHM) intra-host transport to achieve sub-100 μs message transfer latencies for high-bandwidth RGB-D point clouds.

- **Autonomous Navigation Stack (WP-CS2):** Implements the Nav2 framework [16]. Planar range data from the Slamtec RPLiDAR A1M8 (360° FoV, 12 m range) is fused with wheel odometry and torso IMU data using an Extended Kalman Filter (EKF) via the `robot_localization` node. Nav2 manages layered 2D costmaps, obstacle inflation, and dynamic obstacle avoidance using the Regulated Pure Pursuit (RPP) controller.
- **Dual-Arm Manipulation Planning (WP-CS3):** The MoveIt 2 framework [14] manages kinematic planning for the dual 6-DoF arms and waist axis. The kinematic models enforce joint limits, velocity bounds, and dynamic obstacle clearance. Self-collision avoidance matrices are precomputed across the robot URDF model to guarantee that arm-to-arm and arm-to-torso collision-free paths are generated within ≤ 5 ms.
- **Standardized Task Port Layer (WP-CS4):** Encapsulates all robotic capabilities into standardized ROS 2 Action Servers. The application layer commands actions via high-level goal definitions without access to raw motor registers or microcontroller buses, enforcing strict interface decoupling.
- **Perception, Speech, and Foundation AI (WP-CS5):** The perception pipeline performs point cloud filtering, plane extraction, and Euclidean cluster extraction for tabletop object segmentation. Real-time automatic speech recognition is performed locally using an optimized INT8 quantized Whisper model. For autonomous manipulation, the OpenVLA foundation model [11] processes head camera RGB frames and natural language instructions to predict 7-DoF end-effector delta action sequences.
- **Web Teleoperation Dashboard (WP-CS6):** A responsive web application developed in React communicates with ROS 2 via `rosbridge_suite` WebSockets. Low-latency H.264 video streams from the RealSense camera are relayed via WebRTC, enabling real-time remote monitoring, emergency stop triggering, and manual teleoperation.

7.3 Cross-Domain Interface Management

In complex mechatronic systems, physical and operational failures predominantly manifest at domain boundaries. Rigorous cross-domain interface management is therefore essential to prevent integration bottlenecks.

7.3.1 Physical Mounting Standards

Mechanical interfaces across subsystems conform to established international robotics standards:

- **End-Effector Flange:** Both wrist tool mounting plates adhere to the ISO 9409-1-50-4-M6 standard (50 mm pitch circle diameter, 4x M6 threaded holes, and $\varnothing 31.5$ mm locating boss). This standardized interface permits rapid swapping between parallel-jaw grippers, suction cups, and custom inspection tools in under 5 minutes without mechanical redesign.
- **Structural Assembly Fasteners:** All structural connections utilize metric grade 8.8 and 10.9 steel fasteners (DIN 912 socket head cap screws and DIN 7991 countersunk screws) engaging precision-machined threads in aluminum alloy plates or steel threaded inserts (Helicoil / PEM).
- **Subsystem Quick-Disconnects:** The complete upper-body torso mounts to the mobile base chassis using an array of six M6 mounting bolts and two precision locating dowel pins (DIN 6325, $\varnothing 6$ mm m6 tolerance), ensuring sub-0.05 mm positional repeatability upon reassembly.

7.3.2 Electrical Voltage Domains and Power Routing

To isolate high-energy motor commutation noise from sensitive logic and computing circuitry, the power architecture enforces four segregated voltage domains:

1. **Raw Battery Bus (24.0–29.2V, High Power):** Supplies the CubeMars AK70-10 QDD motor drives (shoulder, elbow, waist) and the dual 100 W differential base drive motors. Protected by a 40 A main fuse and safety contactor.
2. **Regulated 19 V Domain ($19.0\text{V} \pm 1.5\%$, 4.0 A Continuous):** Dedicated exclusively to the NVIDIA Jetson Orin NX compute module, derived via an isolated synchronous buck converter.

3. **Regulated 12 V Domain ($12.0\text{V} \pm 2.0\%$, 6.0 A Continuous):** Powers the Feetech STS3215/STS3032 serial bus servos (wrists, grippers, pan/tilt head) and the Intel RealSense depth sensor.
4. **Regulated 5 V & 3.3 V Domains (Low Power Logic):** Powers the Slamtec RPLiDAR, ReSpeaker microphone array, ESP32 microcontrollers, CAN transceivers, and digital sensor pull-ups.

All subsystem grounds converge at a single central star-ground point located on the main power distribution board, mitigating ground potential differentials and eliminating closed ground loops.

7.3.3 Communication Protocols and Fieldbus Determinism

The communication architecture maps data flows to appropriate protocols based on required bandwidth and latency constraints:

- **High-Speed Cognitive Bus (Gigabit Ethernet / USB 3.0):** Carries high-bandwidth sensory data (uncompressed 1280×720 @ 30 fps RGB-D video streams and raw point clouds) between the Intel RealSense camera and the NVIDIA Jetson Orin NX host.
- **System Fieldbus (CAN 2.0B / TWAI):** Connects the Jetson compute host to the distributed ESP32 microcontrollers and CubeMars AK70-10 motor actuators at 1 Mbps. Motor setpoints and telemetry packets operate at a cyclic rate of 1 kHz with packet jitter constrained below $150\text{ }\mu\text{s}$.
- **Subsystem Device Serial Buses (Half-Duplex Asynchronous TTL):** Distributes position commands and reads position/temperature telemetry from the Feetech STS3215/STS3032 serial servos at 1 Mbps using custom multi-drop single-wire UART transceivers.
- **Sensor Peripheral Buses (SPI / I2C / I2S):** Dedicated local buses connecting microcontrollers to peripheral sensors: SPI at 10 MHz for the BNO085 9-DoF IMU, I2S for high-fidelity digital audio input/output, and GPIO interrupt lines for hardware limit switches.

7.3.4 Safety Interlocks and Fault Isolation

The platform enforces a layered safety architecture in compliance with ISO 13482 personal care robotics standards [5]:

- **Hardware Safety Interlock (Level 0):** A dual-channel, normally closed (NC) hardware emergency-stop loop connects the manual E-Stop pushbuttons (located on the torso shoulder and base rear) directly to the de-energizing coil of the main 40 A power contactor. Depressing the E-Stop physically severs battery power from all motor actuators within ≤ 10 ms. Crucially, the 19 V, 12 V, and 5 V logic rails remain fully energized, allowing the Jetson compute host and microcontrollers to maintain telemetry, capture event logs, and broadcast alarm states.
- **Firmware Watchdog Interlock (Level 1):** Each ESP32 microcontroller executes a hardware watchdog timer and monitors continuous CAN communication heartbeats from the high-level host. If communication drops for longer than 50 ms, the local firmware automatically commands motor zero-velocity braking and disables the PWM gate drives.
- **Software Supervisory Interlock (Level 2):** The Task Port safety supervisor monitors Cartesian end-effector velocities, joint torque limits, and distance-to-human thresholds derived from the LiDAR and depth camera. In accordance with ISO 13482 collaborative speed and separation monitoring, the supervisor scales down platform transit speed ($v_{\text{base}} \leq 0.3$ m/s) when obstacles are within 0.8 m, and initiates a software-controlled deceleration stop if a collision trajectory is projected.

8 Bill of Materials, Economics, and Implementation Roadmap

8.1 Consolidated Bill of Materials (BOM)

A foundational objective of the mechatronic design process is reconciling high-performance mobile manipulation capabilities with strict financial and procurement constraints. Grounded in the comprehensive component quotations and local Egyptian market sourcing analysis documented in the project budget evaluation, Table 6 presents the consolidated Bill of Materials (BOM) for the complete core robotic platform.

Table 6: Consolidated Bill of Materials (BOM) for the core semi-humanoid robotic platform.

Component	Technical Specification	Subsystem	Qty	Unit (EGP)	Total (EGP)	Category
Actuation & Drivetrain						
DC Gearmotor with Encoder	24V, 100W, $\sim 30 : 1$ planetary reduction, optical quadrature	Base	2	7,405	14,810	Actuation
QDD Actuator (AK70-10)	CubeMars AK70-10, 24.8 N · m peak, 10:1 planetary, FOC driver	Torso, Arms	9	22,155	199,395	Actuation
Serial Bus Servo (STS3215)	Feetech STS3215, 12V, 2.9 N · m stall, half-duplex TTL bus	Head, Arms	6	1,480	8,880	Actuation
Serial Bus Servo (STS3032)	Feetech STS3032, compact parallel-jaw drive bus servo	Dual Arms	2	1,075	2,150	Actuation
Sensing & Perception Suite						
2D LiDAR Scanner	Slamtec RPLiDAR A1M8, 360° scan, 12 m range, 5.5–10 Hz	Base	1	6,665	6,665	Sensing
Ultrasonic Range Sensors	HC-SR04 sonar array (4 units), base perimeter obstacle coverage	Base	4	205	820	Sensing
Torso 9-DoF IMU	BNO085-class 9-DoF IMU breakout, SPI/I2C fused orientation	Torso	1	930	930	Sensing
RGB-D Perception Camera	Intel RealSense D435/D455, depth and RGB streams, USB 3.0	Head	1	55,000	55,000	Sensing
Microphone Array	ReSpeaker 2-Mic Pi HAT, far-field voice recognition pickup	Head	1	1,685	1,685	Sensing
Audio Feedback Speaker	2 W low-profile speaker with I2S class-D amplifier	Head	1	620	620	Sensing
Compute, Embedded Control & Custom Electronics						
Dedicated Microcontroller	ESP32-WROOM-32 DevKitC, dual-core 240 MHz, TWAI CAN	Subsystems	5	465	2,325	Compute
Main Compute Module	NVIDIA Jetson Orin NX (16 GB) Module + Carrier Board	Torso	1	100,000	100,000	Compute
Compute Cooling Kit	Active heatsink + PWM blower fan for Orin NX compute bay	Torso	1	1,815	1,815	Compute
Custom Base Controller PCB	Dual H-bridge, encoder buffers, ESP32 header, fabricated	Base	1	3,030	3,030	Compute
Custom Power & Waist PCB	Fused busbars, 3x buck regulators, CAN transceiver	Torso	1	4,715	4,715	Compute
Custom Head Interface PCB	TTL servo level shifters, audio amp, ESP32 carrier	Head	1	1,885	1,885	Compute
Custom Arm Interface PCB	CAN transceiver, TTL buffer, ESP32 header (per arm)	Dual Arms	2	2,155	4,310	Compute
Power, Energy Storage & Electrical Safety						
Main Battery Pack	LiFePO ₄ , 24V, 20Ah (~ 480 Wh), internal 40A BMS	Torso	1	13,935	13,935	Power
Smart Battery Charger	24V LiFePO ₄ profile smart charger, 5 A CC/CV	External	1	2,330	2,330	Power
Main Safety Contactor	24V coil, 40 A continuous rating, hardware E-Stop cutoff	Torso	1	1,815	1,815	Power
Structural Hardware & Mechanical Running Gear						
Primary Drive Wheels	$\varnothing 200$ mm high-traction rubber wheels + motor shaft hubs	Base	2	1,970	3,940	Mechanical
Heavy-Duty Swivel Casters	100 mm polyurethane load-rated casters (dual rear support)	Base	2	930	1,860	Mechanical
Core Platform Total Cost (Base + Torso + Head + Dual Arms)					EGP 432,915	100.0%

8.2 Device-Level Subsystem Cost Distribution

In alignment with the modular device architecture detailed in Section 6, Table 7 details the cost share and financial allocation across each independent sub-assembly.

Table 7: Device-level cost distribution and share of the core platform budget.

Subsystem Device	Functional Scope & Key Components	Total Cost (EGP)	Cost Share (%)
Dual 6-DoF Arms	8x AK70-10 QDDs, 4x STS3215, 2x STS3032 grippers, 2x PCBs, 2x MCUs	190,550	44.0%
Torso & Power	AK70-10 waist, Jetson Orin NX 16GB, 24V 20Ah LiFePO ₄ , PDB, BMS	148,160	34.2%
Perception Head	Intel RealSense RGB-D, 2x STS3215 pan/tilt, mic array, speaker, PCB	62,615	14.5%
Mobile Base	2x 100W DC gearmotors, 2D LiDAR, ultrasonic sensors, wheels, PCB	31,590	7.3%
Core Platform Total	Fully integrated baseline mobile manipulator	EGP 432,915	100.0%
<i>Telescopic Column</i>	<i>Optional modular add-on (24V linear drive, rails, tubes, MCU, PCB)</i>	<i>+EGP 32,920</i>	<i>+7.6%</i>
Grand Total	Platform including optional telescopic column	EGP 465,835	–

8.3 Economic Analysis and Category Breakdown

Decomposing the core platform budget (EGP 432,915) by engineering disciplines reveals the economic drivers of the mechatronic architecture:

- **Actuation and Drives (52.0% / EGP 225,235):** Constitutes the largest single investment, comprising the nine CubeMars AK70-10 QDD modules, eight Feetech serial bus servos, and two base drive gearmotors.
- **Compute and Embedded Control (27.3% / EGP 118,080):** Dominated by the NVIDIA Jetson Orin NX (16 GB) module (EGP 100,000), which supplies the 100 TOPS of INT8 tensor acceleration required for on-device VLA policy inference, alongside the five ESP32 MCUs and four custom populated PCBs.
- **Sensing and Perception Suite (15.2% / EGP 65,720):** Primarily driven by the Intel RealSense depth camera (EGP 55,000), complemented by the Slamtec 2D LiDAR, 9-DoF IMU, ultrasonic sonar sensors, and acoustic hardware.
- **Power, Electrical Safety, and Mechanical Structure (5.5% / EGP 23,880):** Encompasses the 480 Wh LiFePO₄ battery pack, smart charger, 40 A safety contactor, high-traction drive wheels, and swivel casters.

8.3.1 Economic Sizing Rationale: The Hybrid Actuation Strategy

A major architectural innovation of this platform is its tiered hybrid actuation strategy. Conventional industrial collaborative arms (e.g., Franka Emika Panda, Universal Robots UR3) utilize strain-wave harmonic gearings and joint torque sensors at all axes, resulting in arm costs exceeding \$25,000–\$35,000 USD (EGP 1,200,000+). Conversely, hobby-grade multi-servo arms fail to provide sufficient holding torque and exhibit high backlash.

By deploying CubeMars AK70-10 Quasi-Direct Drive actuators (24.8 N · m peak torque, 10:1 planetary gear) at the high-load proximal axes (waist tilt, shoulder pitch/roll/yaw, and elbow pitch), the robot secures intrinsic backdrivability, direct motor current torque estimation, and dynamic compliance where impact hazards are greatest. Concurrently, lightweight Feetech STS3215 serial bus servos (2.9 N · m) are deployed at the distal wrist pitch and roll axes, where gravitational cantilever moments are minimal. This tiered approach reduces total manipulator actuation expenditure to EGP 190,550 (~\$3,900 USD) for dual 6-DoF arms—an approximate 70% cost reduction relative to commercial research manipulators.

8.3.2 Domestic Egyptian Sourcing and Sizing Rationale for the Telescopic Column

All selected electronic components (ESP32 microcontrollers, motor drivers, LiFePO₄ cells, connectors) were chosen from supply lines available directly within the Egyptian domestic distribution market or established regional importers. This insulates the project against foreign exchange volatility and protracted customs clearance delays.

Crucially, the telescopic lift column (detailed in Table 8) was evaluated as a modular upgrade (+EGP 32,920, representing a 7.6% budget expansion) and intentionally excluded from the core baseline build. Kinematic simulation in Section 2 demonstrated that combining the mobile base translation with the active waist tilt joint provides a vertical reaching span of 300–1200 mm, successfully covering standard dining tables (750 mm), kitchen countertops (900 mm), and lower storage shelves without an active prismatic mast. Omitting the column from the baseline platform:

1. Reduces dry structural mass by 8.5 kg, lowering the system center of mass by 180 mm and enhancing dynamic tipping margins during rapid mobile base deceleration.
2. Eliminates mechanical binding and backlash risks inherent to sliding prismatic guide tubes under off-axis arm loads.
3. Optimizes fiscal allocation, channeling available budget into high-performance compute (Jetson Orin NX 16GB) and high-fidelity depth perception (Intel RealSense).

Table 8: Detailed Bill of Materials for the optional modular telescopic column add-on.

Component	Specification	Qty	Unit (EGP)	Total (EGP)	Category
Heavy-Duty Linear Drive	24V electric linear actuator, 300 mm stroke, 1500 N thrust	1	17,510	17,510	Actuation
Linear Guide Rail & Carriage	Dual MGN15 recirculating ball rails + carriages	1	6,215	6,215	Mechanical
Telescoping Tube Set	Nested precision aluminum square tubes with bronze bushings	1	7,250	7,250	Mechanical
Dedicated Microcontroller	ESP32-WROOM-32 DevKitC (independent lift controller)	1	465	465	Compute
Custom Column Driver PCB	Populated PCB with H-bridge power stage & limit headers	1	1,480	1,480	Compute
Optional Telescopic Column Subsystem Total Cost				+EGP 32,920	+7.6%

8.4 Risk Register and Mitigation Matrix

The development of a multi-DoF semi-humanoid robot entails technical, operational, and supply chain risks. Table 9 defines the project risk register, evaluated using a standard Risk Priority Number ($RPN = \text{Likelihood [1–5]} \times \text{Severity [1–5]}$) alongside concrete engineering mitigation strategies.

8.5 Phased Implementation Roadmap

The engineering lifecycle progresses across four synchronized phases spanning September through June, structured according to the VDI 2206 macro-cycle model and project milestones.

8.5.1 Phase 1: Component Benchtop Characterization & PCB Prototyping (Months 1–3: Sep–Nov)

The primary objective of Phase 1 is the empirical verification of physical components and custom electronics prior to chassis fabrication:

Table 9: Mechatronic risk register and engineering mitigation matrix.

ID	Risk Description & Operational Impact	Sev (S)	Lik (L)	RPN	Concrete Technical Mitigation Strategy
R1	Import Delays & Currency Fluctuations: Custom electronic components delayed at international shipping ports.	4	3	12	Dual-source component procurement; domestic Egyptian stocking for all passives, MCUs, and power hardware; modular PCB footprints accommodating pin-compatible alternative ICs.
R2	Actuator Thermal Heating: Continuous static holding torques on arm/waist QDDs cause thermal throttling or winding damage.	4	3	12	High-conductivity 6061-T6 aluminum heatsink motor brackets; active convective fan ducting inside torso; software gravity-compensation feeds in MoveIt 2; automatic firmware thermal derating at $T \geq 70^\circ\text{C}$.
R3	CAN Bus Packet Jitter & Saturation: 1 kHz motor control frames collide with sensor telemetry on shared bus, causing control instability.	4	2	8	Dual physically separated CAN 2.0B buses (Bus 1: Torso & Arm QDDs; Bus 2: Base Drivetrain); strict monotonic arbitration ID prioritization; pre-allocated static DMA ring buffers in ESP32 TWAI drivers.
R4	VLA Policy Hallucination: Vision-Language-Action neural policy outputs unconstrained out-of-distribution trajectories.	5	3	15	The Task Port safety supervisor intercepts all VLA action chunks; enforces hard Cartesian bounding boxes, maximum joint speed clamps (≤ 0.5 rad/s), MoveIt 2 collision checks, and impedance contact force abort thresholds.
R5	Battery Rail Transient Voltage Sag: Simultaneous acceleration of dual arms and mobile base draws high surge currents, resetting MCUs.	3	3	9	High-discharge LiFePO ₄ chemistry (2C continuous, 5C pulse); low-ESR bulk electrolytic capacitor banks ($3 \times 2200 \mu\text{F}$) on custom PDB; galvanically isolated DC-DC converters for logic rails (19V, 5V, 3.3V).
R6	Perception Occlusion & Lighting Variation: Tabletop objects occluded by manipulator arms or distorted by intense sunlight.	3	3	9	Active 2-DoF head pan-tilt visual search patterns; multi-frame point cloud temporal filtering; active IR stereo projection from Intel RealSense; complementary ultrasonic proximity detection around base perimeter.

- **Actuator Dynamometer Testing:** Benchtop characterization of the CubeMars AK70-10 QDD actuators. Validate torque constant (K_t), continuous holding thermal limits, and step response bandwidth using a dedicated reaction-torque loadcell fixture.
- **PCB Layout, Fabrication, and Assembly:** Design, review, and fab submission of the four custom circuit boards in KiCad. Assemble surface-mount components, verify rail voltage regulation, and test the 40 A safety contactor cutoff timing (≤ 10 ms).
- **Low-Level Embedded Firmware Bring-Up:** Flash FreeRTOS firmware to the ESP32-WROOM-32 microcontrollers. Establish deterministic 1 kHz TWAI CAN communication with motor inverters and 1 Mbps half-duplex TTL serial bus control with Feitech servos.
- **Milestone 1 Exit Criteria:** Fully operational benchtop hardware loop: Jetson Orin NX commanding synchronized multi-axis motor motions over CAN and TTL buses with zero dropped packets over 1 hour of continuous cycling.

8.5.2 Phase 2: Software-in-the-Loop (SIL) and Digital Twin Simulation (Months 3–5: Dec–Jan)

Operating concurrently with the physical manufacturing track, Phase 2 constructs a physics-accurate digital twin to validate high-level software stacks:

- **Digital Twin Simulation (Isaac Sim / Gazebo):** Formulate the comprehensive robot URDF/Xacro model, incorporating exact link masses, centers of gravity, and inertia tensors derived from CAD solid models. Import the digital twin into NVIDIA Isaac Sim [17] and Gazebo Harmonic.
- **Autonomous Navigation Stack Validation:** Deploy the Nav2 stack within a simulated multi-room domestic apartment modeled after the RoboCup@Home arena. Validate 2D LiDAR SLAM, costmap inflation tuning, dynamic obstacle clearance, and smooth trajectory tracking.
- **MoveIt 2 Manipulation Planning:** Configure MoveIt 2 kinematic solvers for the dual 6-DoF arms and waist axis. Verify collision-free motion planning, self-collision matrices, and Cartesian tool-center-point (TCP) trajectory interpolation.
- **Parallel Manufacturing Track:** Execution of chassis CNC plate machining, waterjet cutting, 3D printing of structural housings (MJF PA12), and precision welding of the torso space-frame.
- **Milestone 2 Exit Criteria:** Complete autonomous pick-and-place pipeline validated in simulation (Nav2 navigation + RealSense segmentation + MoveIt 2 trajectory planning) achieving $> 90\%$ simulated task success rate.

8.5.3 Phase 3: Physical Assembly & Hardware-in-the-Loop (HIL) Integration (Months 5–7: Feb–Mar)

Phase 3 integrates the manufactured mechanical structure, custom electronics, and low-level firmware into a fully operational physical robot:

- **Mechanical Assembly and Harnessing:** Assemble the mobile base drivetrain, bolt the torso space-frame to the chassis with precision dowel pins, mount the dual 6-DoF arms, and install the 2-DoF sensor head. Route twisted-pair shielded wire harnesses through cable carriers and slip rings.
- **Power Distribution and Safety Commissioning:** Install the 24V, 20Ah LiFePO₄ battery pack, smart BMS, and custom PDB. Execute full-system electrical isolation testing and verify physical E-Stop contactor cutoff under live motor load.
- **Hardware-in-the-Loop Controller Tuning:** Connect the Jetson Orin NX host to the physical microcontroller fieldbus via micro-ROS. Calibrate joint zero-positions, tune closed-loop velocity PID gains on the mobile base, and configure gravity-compensation feedforward tables on arm joints.
- **Web Teleoperation Dashboard Bring-Up:** Deploy the web dashboard on the local network. Establish low-latency WebRTC video streaming from the RealSense camera and verify manual Cartesian jog teleoperation.
- **Milestone 3 Exit Criteria:** Physical platform standing under its own power, executing synchronized dual-arm motions and mobile base teleoperation with smooth, vibration-free motion and verified emergency stop functionality.

8.5.4 Phase 4: Task Port Validation & Proof-of-Concept Demonstration (Months 7–9: Apr–Jun)

The concluding phase verifies the core architectural premise of the platform: decoupling capability from intent via the standardized Task Port interface:

- **Task Port Action Server Deployment:** Expose standardized ROS 2 Action Servers for navigation, arm manipulation, and speech interaction (`/navigate_to_pose`, `/pick_object`, `/place_object`, and `/speak_text`).
- **Empirical Proof-of-Concept Task Execution:** Flash an external, task-specific application node representing the canonical RoboCup@Home General Purpose Service Robot (GPSR) Category I task:
 1. System accepts natural language command: *“Robot, navigate to the dining table, pick up the beverage can, and place it on the counter.”*
 2. Base autonomously navigates 6 m through a furnished domestic environment, clearing dynamic obstacles using Nav2.
 3. Perception head executes active 3D point cloud segmentation via RealSense, identifying target object coordinates and surface grasp orientation.
 4. Left manipulator executes compliant collision-free reach, secures workpiece with parallel gripper, and verifies grasp via motor current feedback.
 5. Platform navigates to destination countertop, executes controlled placement, and provides spoken verbal completion confirmation.
- **Validation Against Engineering Requirements:** Formally benchmark platform performance against the engineering requirements list (Table 3), recording transit velocity (0.9 m/s), payload capacity (2.2 kg), Cartesian repeatability (± 1.4 mm), and continuous battery runtime (4.6 h).
- **Embodied AI Exploration (Stretch Goal):** Evaluate zero-shot trajectory inference using an INT8 quantized OpenVLA foundation model [11] for tabletop object sorting.
- **Milestone 4 Exit Criteria:** Successful end-to-end physical execution of the autonomous pick-and-place task, verified compliance with all Fixed Requirements (FF), submission of final engineering documentation, and public graduation project defense.

9 Conclusion and Research Outlook

9.1 Conceptual Design Synthesis

This report has established the conceptual mechatronic system design and foundational architecture for a modular, general-purpose semi-humanoid robotic platform. Grounded in the VDI 2206 design methodology for mechatronic and cyber-physical systems [6], the platform architecture systematically resolves the classical trade-off in service and collaborative robotics: uniting the energy efficiency, continuous stability, and payload capacity of wheeled autonomous mobile robots (AMRs) with the anthropomorphic reach, bimanual coordination, and social compliance of dual-arm manipulators [3, 1].

By grounding the functional requirements in standardized international service robotics benchmarks—specifically the RoboCup@Home General Purpose Service Robot (GPSR), Clean Up, and Human-Robot Interaction challenges [7]—the physical body is optimized for unstructured interior spaces. The platform combines a high-traction differential mobile base (520 mm $\times$ 520 mm footprint) capable of traversing standard 750 mm interior doorways, a rigid fixed structural mast torso (500–1350 mm reaching envelope, with modular waist tilt mounting and analyzed optional telescopic column), dual 6-DoF manipulators powered by a tiered Quasi-Direct Drive (QDD) actuation topology, compliant parallel-jaw end effectors, and an expressive 2-DoF stereo-vision perception head, establishing a robust 18-DoF baseline platform.

A core tenet of the design methodology is the complete functional and logical decoupling of physical mechatronic capability from application-specific intent. Through the formalization of the standardized Task Port abstraction layer, higher-level cognitive agents, speech dialogue engines, and modern Vision-Language-Action (VLA) foundation models interact exclusively with high-level action primitives (`/navigate_to_pose`, `/pick_object`) without accessing low-level motor registers. Real-time supervisory safety nodes independently monitor Cartesian velocity bounds, joint torque saturation, and human proximity thresholds in compliance with ISO 13482 personal care robotics safety standards [5], guaranteeing fail-safe physical operation.

Finally, by prioritizing domestic Egyptian market component availability and engineering a hybrid actuation strategy that deploys CubeMars AK70-10 QDD modules at high-load proximal joints and low-cost Feetech serial bus servos at distal joints, the platform achieves high dynamic performance at a baseline Bill of Materials cost of EGP 432,915 (~\$8,900 USD). This represents an order-of-magnitude cost reduction relative to commercial research platforms, establishing an accessible, publication-grade foundation for embodied AI and mobile manipulation research.

9.2 Key Research Contributions

The conceptual system design presented in this document yields three primary contributions to mechatronic systems engineering and robotics literature:

1. **Decoupled Task Port Abstraction for Embodied AI:** The report formalizes an architectural boundary that bridges the profound operational bandwidth mismatch between high-level foundation AI policies and hard real-time motor commutation. By placing real-time trajectory interpolators (1 kHz) and a deterministic ISO 13482 safety supervisor between 10 Hz probabilistic VLA action chunk streams and distributed motor fieldbuses, the Task Port eliminates structural vibration hazards and isolates physical hardware from neural policy hallucinations.
2. **Cost-Optimized Tiered Hybrid Actuation Topology:** Rather than incurring the prohibitive financial penalties of all-harmonic collaborative arms or the fragile performance of hobby servo mechanisms, the platform proves the viability of a tiered QDD-plus-bus-servo topology. Positioning high-torque Quasi-Direct Drive actuators at proximal shoulder and elbow axes (with modular waist allocation) provides intrinsic backdrivability, impact absorption, and open-loop torque regulation where dynamic interaction loads dominate, while distal serial bus servos drastically lower moving arm inertia and system cost.
3. **VDI 2206-Compliant Cyber-Physical Hardware Modularity:** Decomposing the platform into four modular, independently testable hardware devices (Base, Torso/Power, Head, and Dual Arms) interconnected via four custom populated PCBs and deterministic CAN buses eliminates monolithic wiring harnesses, isolates high-current motor switching noise, and establishes an upgradeable foundation that accommodates optional add-ons, such as the analyzed 300 mm prismatic lift column.

9.3 Research Outlook and Future Trajectories

With the conceptual design and domain allocations locked, future research following physical prototyping will investigate three advanced engineering trajectories:

9.3.1 Whole-Body Operational Space Control (WBC)

In the current baseline architecture, base navigation and dual-arm manipulation are coordinated sequentially through the Task Port. Future research will implement unified Whole-Body Operational Space Control (WBC) formulated as a hierarchical Quadratic Program (QP) [18, 19]. Under this framework, base planar translation ($SE(2)$) and dual-arm end-effector trajectories ($SE(3)$) are resolved simultaneously across a 14-DoF unified kinematic manifold (expandable to 15-DoF with active waist tilt):

$$\min_{\mathbf{\ddot{q}}, \boldsymbol{\tau}} \quad \frac{1}{2} \|\mathbf{J}_{\text{task}} \mathbf{\ddot{q}} - \ddot{\mathbf{x}}_{\text{des}}\|_{\mathbf{W}_x}^2 + \frac{1}{2} \|\boldsymbol{\tau}\|_{\mathbf{W}_\tau}^2 \quad (39)$$

subject to:

$$\mathbf{M}(\mathbf{q}) \mathbf{\ddot{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau} + \mathbf{J}_c^T \mathbf{F}_c \quad (40)$$

$$\mathbf{q}_{\min} \leq \mathbf{q} \leq \mathbf{q}_{\max}, \quad \dot{\mathbf{q}}_{\min} \leq \dot{\mathbf{q}} \leq \dot{\mathbf{q}}_{\max}, \quad \boldsymbol{\tau}_{\min} \leq \boldsymbol{\tau} \leq \boldsymbol{\tau}_{\max} \quad (41)$$

$$\mathbf{p}_{\text{ZMP}}(\ddot{\mathbf{q}}, \boldsymbol{\tau}) \in \mathcal{S}_{\text{support}} \quad (42)$$

Enforcing Zero-Moment Point (ZMP) and Centroidal Momentum Matrix (CMM) constraints ensures that aggressive mobile base acceleration maneuvers and dynamic dual-arm reaching loads never breach the physical tipping boundaries of the wheel support polygon $\mathcal{S}_{\text{support}}$.

9.3.2 Sim-to-Real Reinforcement Learning and Foundation Policy Distillation

Exploiting the digital twin constructed in NVIDIA Isaac Sim / Isaac Lab [17], future work will explore large-scale simulation-to-reality (sim-to-real) transfer [20, 21]. By executing thousands of parallelized simulation environments on GPU clusters, reinforcement learning policies will be trained for complex contact-rich manipulation tasks, including opening spring-loaded domestic doors, unscrewing jar lids, and compliant dual-arm sheet folding. Massively parallelized Domain Randomization (DR) will be injected across link inertial properties, joint Coulomb and viscous friction, motor torque constants, communication bus latencies, and photorealistic ambient lighting conditions.

Concurrently, the onboard NVIDIA Jetson Orin NX host will support parameter-efficient fine-tuning (LoRA) of OpenVLA and RT-X foundation models [11, 12] directly on teleoperation demonstration datasets captured via the web dashboard, investigating closed-loop visual imitation learning under tight computational constraints.

9.3.3 Multi-Agent Fleet Orchestration and Cloud-Native Robotics

Finally, the standardized Task Port interface establishes the logical foundation for multi-robot collaborative fleet operations. Future developments will integrate the Eclipse Zenoh networking protocol and cloud-native robotics platforms [22]. This will enable distributed semantic spatial mapping, where multiple semi-humanoid platforms collaboratively update a centralized occupancy and 3D Gaussian splatting representation of large public facilities (e.g., healthcare clinics, university campuses, smart logistics warehouses). Furthermore, peer-to-peer Task Port synchronization will facilitate cooperative bimanual manipulation, enabling two robots to coordinate base navigation and arm grasping to transport bulky, non-rigid payloads exceeding the physical capacity of a single platform.

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